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# Universal Plastic Shredder V2.0

## *Electrical Control & Power System*

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PORTLAND STATE UNIVERSITY  
MASEEH COLLEGE OF ENGINEERING & COMPUTER SCIENCE  
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

*Final Report — Rough Draft*

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*[NOTE: This is a rough draft. Sections marked TODO, NOTE, or QUESTION indicate areas where we need to expand content, gather data, or get advisor input before the final submission. Bullet points are placeholders for fuller prose in the final version.]*

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# 1 Executive Summary

- Ichor Systems generates significant plastic waste (PVC, polypropylene, LDPE) during semiconductor equipment manufacturing, currently sent to landfill.
- Commercial industrial shredders cost \$10,000+ and consumer units cannot handle the thick rigid sheets (0.25–0.5 in).
- Our project designs the electrical control and power system for a custom industrial-grade plastic shredder, capable of shredding into 1-inch pieces.
- Key elements: VFD-controlled 3 HP motor at 70–90 RPM output, two-hand deadman control, hardwired E-stop, jam-detection auto-recovery with 3-strike lockout, NEMA-enclosed PLC and HMI.
- Total budget: \$3,000 (EE portion approximately \$1,000).

[QUESTION: Should the executive summary include high-level results / status (e.g., “shredder validated to X HP load with Y throughput rate”) once we complete testing? Currently we have signal-chain validation but not full mechanical load testing.]

# 2 Introduction

## 2.1 Background and Motivation

- Ichor Systems is a Portland-area semiconductor equipment manufacturer.
- Their manufacturing processes produce substantial volumes of rigid plastic sheet waste.
- Existing disposal route: landfill – environmentally and economically suboptimal.
- A previous 2023–2024 PSU capstone team built a 2 HP household shredder for the EPL. This project scales to industrial-grade.

## 2.2 Project Scope

- EE team responsibility: motor control, safety circuits, PLC logic, HMI, power distribution.
- ME team (separate) handles mechanical frame, blade assembly, gearbox.

- Integration plan: mechanical structure delivered with motor mount; EE team commissions controls.

[TODO: Add a one-paragraph background on dynamic recycling / circular economy framing for the introduction.]

## 3 Requirements

### 3.1 Must-Have

- Shred 0.25–0.5 in plastic sheet into approximately 1-inch pieces.
- Controllable VFD supporting 3–5 HP motor at 70–90 RPM output shaft speed.
- Follow NFPA 70 (NEC) electrical design guidance.
- Include emergency stop, overcurrent/overload protection, at least one safety interlock.
- Include an HMI for operator control and status display.
- Stay within \$3,000 total project budget.

### 3.2 Should-Have

- Clear status and fault indicator lights.
- Follow NEMA enclosure guidelines.
- Clearly labeled controls.

### 3.3 May-Have (if time/budget allow)

- Wireless monitoring (read-only).
- Data logging of operating hours and faults.
- Additional safety sensing (capacitive touch, thermal).

## 4 System Architecture

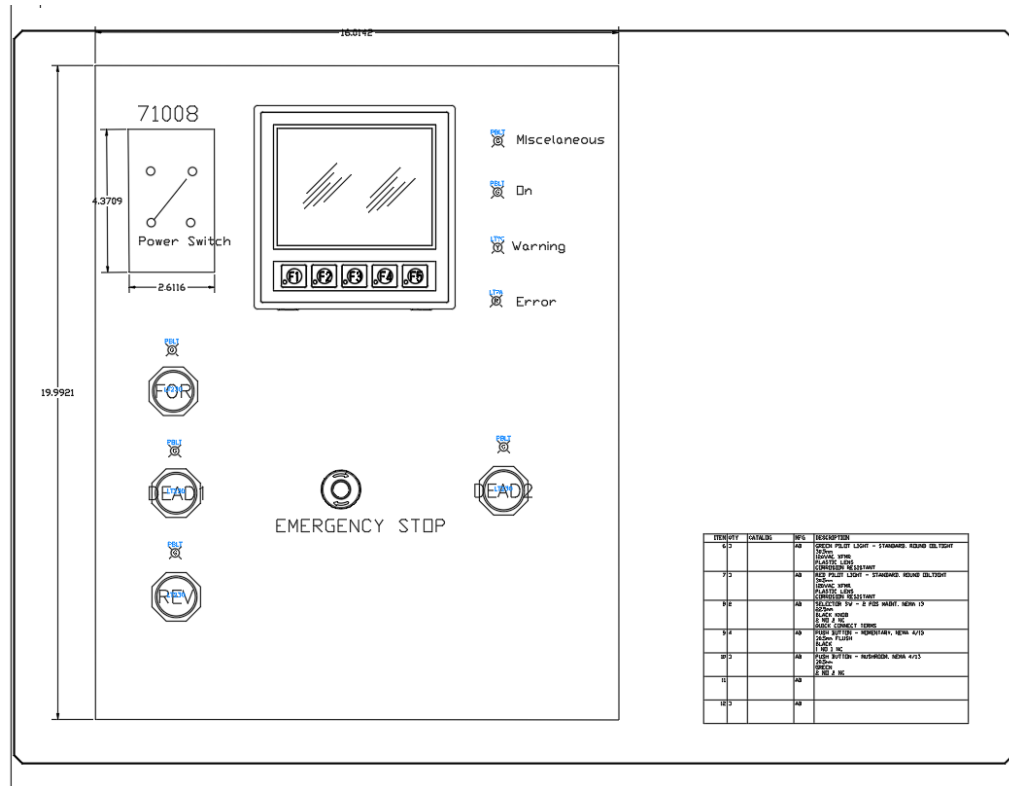
### 4.1 Hardware Subsystems

- **AC power input:** 120 VAC mains → main disconnect → branch breakers → VFD and 24V PSU.

- **VFD:** Huanyang HY02D211B-T (2.2 kW, 110V single-phase input, 220V 3-phase output boost-type). Controlled via FOR/REV/RST digital inputs.
- **Motor:** bench/test unit is a GE 5KE182BC205B (3 HP, 230 V low-voltage connection, 1750 RPM, FLA 7.6 A); its nameplate drives the VFD motor parameters. **[TODO: Final production motor selection is TBD (3–5 HP per requirements).]**
- **Brake resistor:** 75  $\Omega$ , 500 W, connected to VFD's P/Pr terminals to handle regenerative energy during deceleration and reversal.
- **PLC:** AutomationDirect H2-DM1E (Do-more CPU) on DL205 base, D2-08ND3 discrete input module.
- **HMI:** AutomationDirect C-more Micro-Graphic EA1-T4CL 4-inch color touch panel, connected to the H2-DM1E CPU serial communication port (operator commands to the PLC; status/data back to the panel). Powered from the shared Mean Well NDR-480-24 24 VDC supply.
- **Safety circuit:** Hardwired E-stop loop and two-hand deadman start; motor contactor energized through the safety chain. Access guarding at the feed is handled by the ME team's mechanical design, so no separate electrical lid interlock is used.
- **24V DC supply:** Mean Well NDR-480-24 (480 W) for control voltage.

## 4.2 Signal Flow

- Operator inputs (deadman buttons, E-stop, reset)  $\rightarrow$  PLC discrete inputs (D2-08ND3, sourcing mode).
- PLC outputs (D2-08TR relay module, common to PSU  $-V$ )  $\rightarrow$  VFD FOR (Y7), REV (Y6), RST (Y5) inputs in sinking (NPN) mode.
- VFD fault relay (FC + FA, NO active-high)  $\rightarrow$  PLC X5 input. X5 = HIGH when VFD trips on fault, LOW when healthy.
- HMI  $\leftrightarrow$  PLC via serial (H2-DM1E CPU serial COM port; matched baud, parity, and stop bits on both ends).



**Figure 1:** Operator control panel layout: HMI, power switch, forward/reverse and deadman pushbuttons, emergency stop, and status indicators.

[TODO: Add a system block diagram (power + signal flow) to complement the panel layout above.]

## 5 Hardware Design

### 5.1 Power Distribution

- Main disconnect / power switch: PowerTec 71008 two-pole (DPST) START/STOP switch, 120–230 VAC single-phase, rated 20 A / 3 HP max; switches both the line and neutral conductors.
- VFD branch breaker: 30 A C-curve DIN-rail (handles inrush, sized per NEC 430.52).
- 24V PSU branch breaker: 10 A C-curve DIN-rail.
- **Conductors:** power runs use 12 AWG copper (THHN/MTW), rated 20 A / 25 A / 30 A at the 60/75/90°C ampacity columns of NEC Table 310.16 – well above the motor’s 7.6 A FLA and the VFD input current. Control wiring uses 16–18 AWG,



landed directly as straight stripped conductors at the terminal blocks (no ferrules used in this build).

- **Fuse / OCPD sizing:** branch and component fuses are sized at **125% of the protected component's rated continuous load** per NEC 210.20(A) / 215.3. (The motor branch breaker is the exception – NEC 430.52 permits a higher multiple of FLC to ride through inrush; running overload is handled by the VFD's electronic motor-overload function.)
- All branches fused individually at terminal blocks for downstream loads.

## 5.2 Motor and VFD

The Huanyang HY02D211B-T was selected over alternatives via a decision matrix (price, documentation, reliability). A brake resistor on the P/Pr terminals absorbs the regenerative energy from rapid reversal during the jam-clear cycle. The drive is programmed as shown in Table 1.

**Table 1:** Programmed VFD parameters (Huanyang HY02D211B-T).

| Param | What it does              | Value    | Why this value                                                                                       |
|-------|---------------------------|----------|------------------------------------------------------------------------------------------------------|
| PD141 | Motor rated voltage       | 230 V    | Matches the GE test-motor nameplate (230 V low-voltage connection); sets the base of the V/Hz curve. |
| PD142 | Motor rated current (FLA) | 7.6 A    | Nameplate full-load amps; sets the drive's current limit and electronic-overload reference.          |
| PD143 | Number of motor poles     | 4        | Nameplate; 4-pole = 1800 RPM synchronous, so the drive resolves speed and slip correctly.            |
| PD144 | Motor rated speed         | 1750 RPM | Nameplate full-load speed; with the pole count it lets the drive estimate slip.                      |

*continued on next page...*

| Param | What it does                  | Value        | Why this value                                                                                                                     |
|-------|-------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------|
| PD002 | Run / frequency source        | 1 (external) | Speed comes from the external 0–10 V analog input (VI) driven by the PLC, not the keypad.                                          |
| PD052 | Relay output (FA/FC) function | 02 (fault)   | Closes the FA–FC relay on any fault so the PLC reads a jam/trip on input X5.                                                       |
| PD054 | Analog output (VO) function   | 1 (current)  | Outputs 0–10 V proportional to motor current to the PLC analog input for monitoring.                                               |
| PD123 | Over-torque action            | 3 (trip)     | Over-torque self-trips the drive (which fires the fault relay); see the note below.                                                |
| PD124 | Over-torque threshold         | 150%         | Set between no-load current ( $\sim 0.5$ A) and measured jam current ( $> 11$ A) so normal shredding does not trip but a jam does. |
| PD125 | Over-torque detection time    | 3.0 s        | The over-torque must persist 3 s before tripping, so brief load spikes (feeding a piece) do not nuisance-trip.                     |

*[NOTE: The over-torque relay function (PD052 = 12) did not fire the relay in our specific Huanyang revision when PD123 = 2 (continue running). Documented as a firmware-specific limitation (see Table 1). Working configuration uses PD052 = 02 (fault indication) and PD123 = 3 (over-torque triggers self-trip, which fires the fault relay). The PLC then runs the unjam routine including a RST pulse to clear the VFD’s latched fault state between retry attempts.]*

### 5.3 Safety Circuit

- E-stop: NC contact (E22B1) hardwired in series with the SD-N35 contactor coil. Pressing it drops the coil and opens the contactor, removing motor power independently of

the PLC.

- Two-hand deadman: both buttons must be held continuously to enable the motor (evaluated by the PLC; see the Safety Functions assessment below).
- Access guarding: handled by the ME team's mechanical design (feed geometry), so no separate electrical lid interlock is used (this was a may-have, not a requirement).
- Motor contactor: the SD-N35 is energized through the hardwired safety chain, and the VFD is powered through it, so the safety chain has hardware precedence over the PLC.

[TODO: Add safety circuit schematic. Reference NFPA 79 section 9 for safety-related stop functions.]

## 5.4 Jam Detection Signal Chain

- VFD's multi-function relay (terminals FA, FB, FC) programmed for Fault Indication (PD052 = 02). **Only FA and FC are used** for jam detection; the FB (normally-closed) contact is left unconnected.
- Wired as an NO active-high pair: +24V (through a 500 mA fast-blow fuse) feeds the FA contact; the FC common returns to PLC X5 input on D2-08ND3 (Slot 2). When the VFD trips, FA-FC closes and +24V reaches X5.
- PLC sees X5 = LOW during normal operation, X5 = HIGH when VFD trips on any fault (over-torque, overvoltage, overheat, etc.).
- PLC ladder reads [ X5 ] on rising edge to detect new fault events.
- Wire-break failure mode is handled at the safety-circuit level by the independent hard-wired E-stop loop and motor contactor, rather than at the jam-signal level.
- Full terminal-by-terminal wiring is given in the I/O point list (Appendix B) and the VFD parameter/wiring table (Appendix D).

# 6 Software Design

## 6.1 PLC Ladder Logic Architecture

The control program runs on the H2-DM1E CPU in the DirectLOGIC 205 D2-09B-1 rack and was written in Do-more Designer. The program is 31 rungs long and is exported in `electrical/Circuit_Logic.pdf`. It implements a state machine for normal operation plus a 3-strike jam recovery sequence and a hard-lockout path.

## Key Terms

| Tag                    | Meaning                     | Tag                       | Meaning                             |
|------------------------|-----------------------------|---------------------------|-------------------------------------|
| FORWARDS<br>(X0)       | Forward pushbutton          | STOP_MOTOR<br>(C15)       | Hard stop / lockout flag            |
| REVERSE<br>(X1)        | Reverse pushbutton          | PAUSE_MOTOR<br>(C17)      | Post-reverse coast-to-stop          |
| DEADMAN_A_OK<br>(X2)   | Deadman button A            | ERROR_RESET<br>(C18)      | Fault reset coil                    |
| DEADMAN_B_OK<br>(X3)   | Deadman button B            | JAMMED_MOTOR<br>(C20)     | 3-strike lockout flag               |
| STOP_PB<br>(X4)        | E-stop NC contact monitor   | CTSL_TRIG<br>(C21)        | Cap-touch trigger flag              |
| OVERCURRENT<br>(X5)    | VFD fault relay input       | CHANGE_SCREEN<br>(C22)    | HMI screen-change trigger           |
| CAP_TOUCH_TRIP<br>(X7) | Cap-touch safety input      | TESTMAN_A_OK<br>(C23)     | HMI soft deadman A                  |
| FWD_LED<br>(Y0)        | Forward indicator LED       | TESTMAN_B_OK<br>(C24)     | HMI soft deadman B                  |
| REVERSE_LED<br>(Y1)    | Reverse indicator LED       | OVERCURRENT_TEST<br>(C25) | HMI soft overcurrent test           |
| DMAN_A_LED<br>(Y2)     | Deadman A indicator LED     | E_STOP_TRIG<br>(C26)      | E-stop triggered flag               |
| DMAN_B_LED<br>(Y3)     | Deadman B indicator LED     | T0–T1                     | Overcurrent / unjam off-delay (2 s) |
| E_LIGHT<br>(Y4)        | Error / fault indicator LED | T2                        | Lockout accumulating tmr (30 s)     |
| VFD_RESET<br>(Y5)      | VFD fault reset output      | T3                        | Post-reverse pause (2 s)            |
| VFD_RUN_FWD<br>(Y6)    | VFD forward run output      | T4                        | HMI screen timeout (3 s)            |
| VFD_RUN_RVS<br>(Y7)    | VFD reverse run output      | CT0                       | Jam-event counter (pre-set 3)       |

*continued on next page...*

| Tag                   | Meaning                    |  | Tag               | Meaning                      |
|-----------------------|----------------------------|--|-------------------|------------------------------|
| DEADMAN_OK<br>(C0)    | Both deadman buttons held  |  | V100              | HMI screen selector          |
| HMI_STP_REQ<br>(C1)   | HMI stop button request    |  | V2000             | Speed setpoint (0–1600)      |
| PERMISSIVE_OK<br>(C3) | Run permissive latch       |  | V3000             | Scaled motor current (0–140) |
| FORWARDS_SET<br>(C8)  | Forward direction latch    |  | WX0 / WY0         | Analog in / out words        |
| REVERSE_SET<br>(C9)   | Reverse direction latch    |  | \$FirstScan, \$On | Do-more system bits          |
| JAM_LATCH<br>(C10)    | Overcurrent jam latch      |  |                   |                              |
| RUN_REQUEST<br>(C12)  | Motor run command latch    |  |                   |                              |
| UNJAM_R<br>(C13)      | Unjam reverse stroke latch |  |                   |                              |

Inputs (X-registers) come from physical pushbuttons, the VFD fault relay, and a capacitive-touch safety sensor. C-registers hold latched internal states. Y outputs drive panel LEDs and the VFD's run / reset terminals. The full address-by-address reference — with electrical types, source modules, and detailed semantics — is reproduced in Appendix F.

### 6.1.1 State Machine

- **IDLE** → **RUNNING** when both deadman buttons held AND not locked out AND startup mask done.
- **RUNNING** → **JAM\_DETECTED** on rising edge of X5 (VFD trip → fault relay fires → X5 goes HIGH).
- **JAM\_DETECTED** → **RST\_PULSE** if jam count < 3, → **LOCKOUT** if jam count ≥ 3.
- **RST\_PULSE** (500 ms) → **REVERSE** (2 s).
- **REVERSE** → **RESUME** (brief delay) → back to **RUNNING**.
- **LOCKOUT** → **IDLE** only via dedicated reset button.

### 6.1.2 Jam Counter Reset Logic

- Counter (CT0) increments on each X5 rising edge (each new VFD fault triggers a jam event).
- Resets when motor runs cleanly for 30 continuous seconds (T2 accumulating timer).
- Also resets via the HMI Screen 3 ERROR\_OFF button (ERROR.RESET, C18).
- Whichever condition fires first wins.

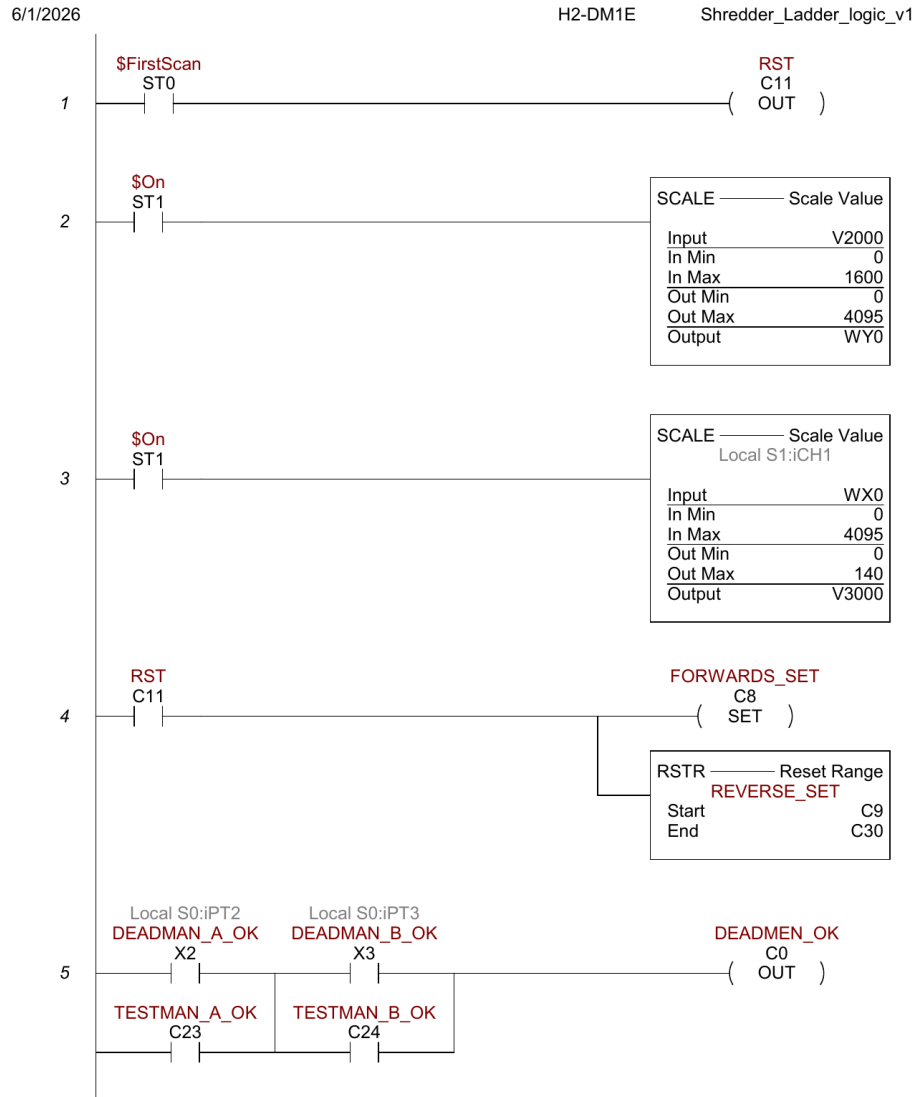
### 6.1.3 Startup Mask

- For first scan after PLC power-up, the FORWARDS\_SET / REVERSE\_SET pair is initialized via rung 4 (defaults to forward direction).
- In the active-high NO X5 wiring, no boot-time mask is required because the relay is naturally open at startup.

### 6.1.4 Rung-by-Rung Description with Ladder Logic Export

The 31 rungs are reproduced below directly from the Do-more Designer export (`electrical/Circuit.Logic.pdf`), with the actual ladder image on each page paired with a description of the rungs it contains. This lets a reviewer see the rung graphic next to the variable map (above) and the narrative below.

## Page 1 of 7 — Rungs 1–5: Power-up Init and Deadman Permissive

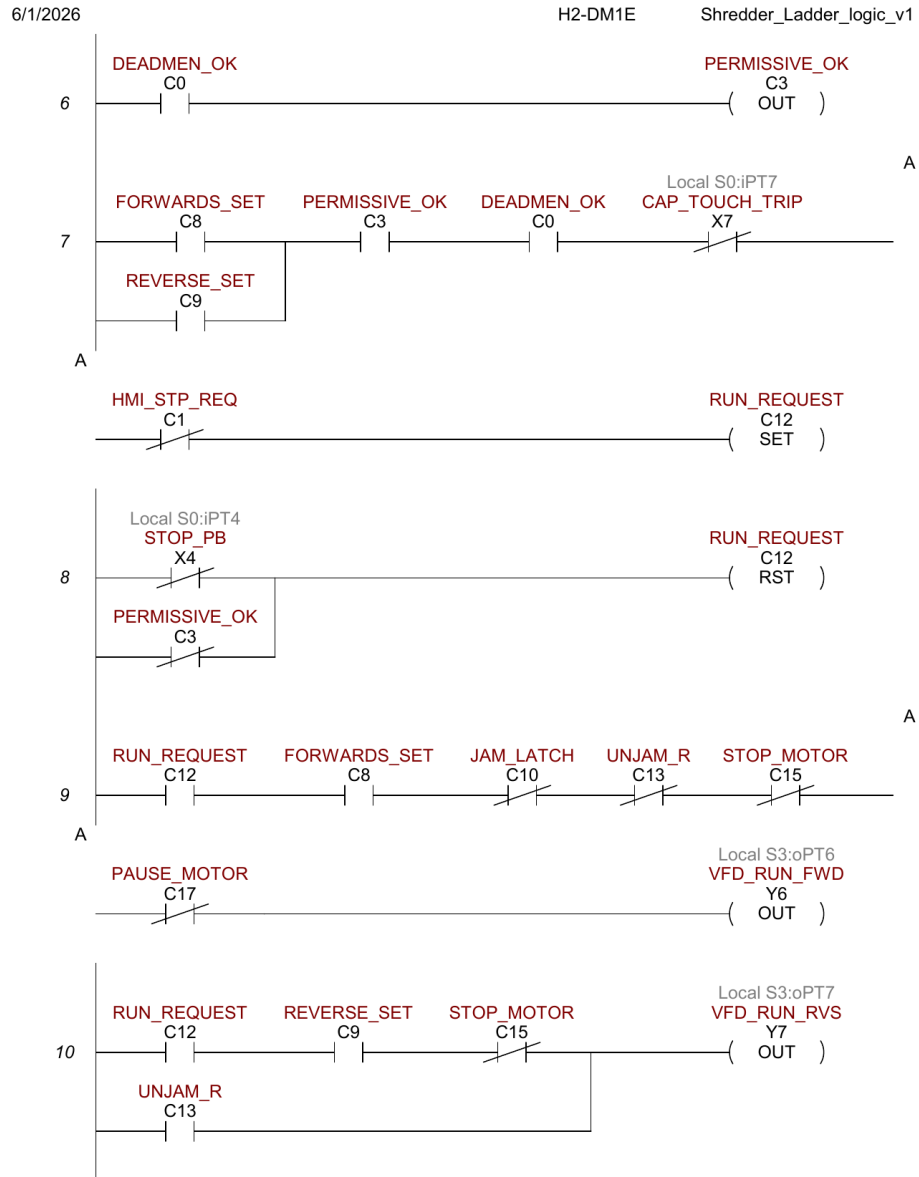


**Figure 2:** Ladder export page 1. Power-up init and deadman permissive (rungs 1–5).

- **Rung 1** (\$FirstScan → RST C11): on the very first scan after power-up, C11 is reset, triggering the init path in rung 4.
- **Rung 2** (\$On → SCALE V2000 [0–1600] → WY0 [0–4095]): every scan, the HMI speed setpoint V2000 is linearly scaled to raw DAC counts for the F2-08DA-2 CH1 → VFD VI.
- **Rung 3** (\$On → SCALE WX0 [0–4095] → V3000 [0–140]): raw ADC from F2-08AD-1 CH1 (VFD VO motor current) is scaled to a 0–140 display value stored in V3000 (CURRENT tag).
- **Rung 4** (NOT C11 → SET FORWARDS\_SET + RSTR C9–C30): initializes FORWARDS\_SET and clears C9–C30 on first power-cycle scan.
- **Rung 5** ((X2 OR C23) AND (X3 OR C24) → OUT C0): computes DEADMEN\_OK by AND-ing the two deadman paths (physical OR HMI-soft test buttons).



## Page 2 of 7 — Rungs 6–10: RUN\_REQUEST Gating and Run Outputs



Page 2

\$Main

**Figure 3:** Ladder export page 2. RUN\_REQUEST gating and run outputs Y6/Y7 (rungs 6–10).

- **Rung 6** ( $C0 \rightarrow \text{OUT } C3$ ): mirrors DEADMEN\_OK to PERMISSIVE\_OK so the permissive logic is referenced separately from the deadman coil.
- **Rung 7** ( $((C8 \text{ OR } C9) \text{ AND } C3 \text{ AND } C0 \text{ AND NOT } X7 \rightarrow \text{SET } C12)$ ): RUN\_REQUEST is set when a direction is selected, permissives are OK, and the capacitive touch is not tripped. The HMI STOP button (NOT C1) is a separate inhibit path.
- **Rung 8** ( $(\text{NOT } X4 \text{ OR NOT } C3 \rightarrow \text{RST } C12)$ ): RUN\_REQUEST is immediately reset if the E-stop opens (X4 LOW) or the deadman permissive drops.
- **Rung 9** ( $C12 \text{ AND } C8 \text{ AND NOT } C10 \text{ AND NOT } C13 \text{ AND NOT } C15 \text{ AND NOT } C17 \rightarrow \text{OUT } Y6$ ): forward run output. Requires RUN\_REQUEST + FORWARDS\_SET + no active jam + not in reverse unjam stroke + no hard stop + post-reverse buffer elapsed.
- **Rung 10** ( $((C12 \text{ AND } C9 \text{ AND NOT } C15) \text{ OR } C13 \rightarrow \text{OUT } Y7)$ ): reverse run has two paths: manual reverse (RUN\_REQUEST AND REVERSE\_SET AND NOT STOP\_MOTOR), or unconditional unjam reverse (UNJAM\_R alone).

## Page 3 of 7 — Rungs 11–16: Indicator LEDs and Direction Latches

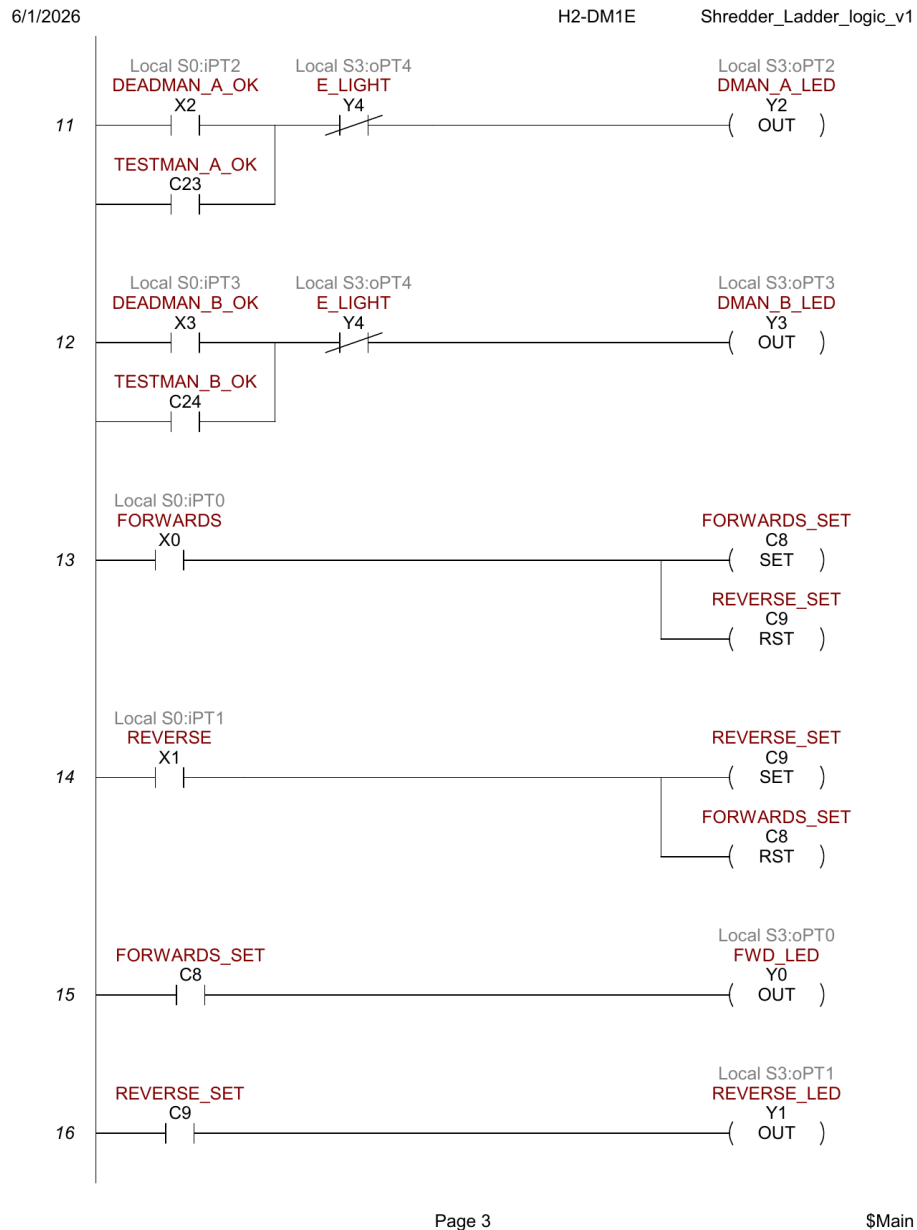
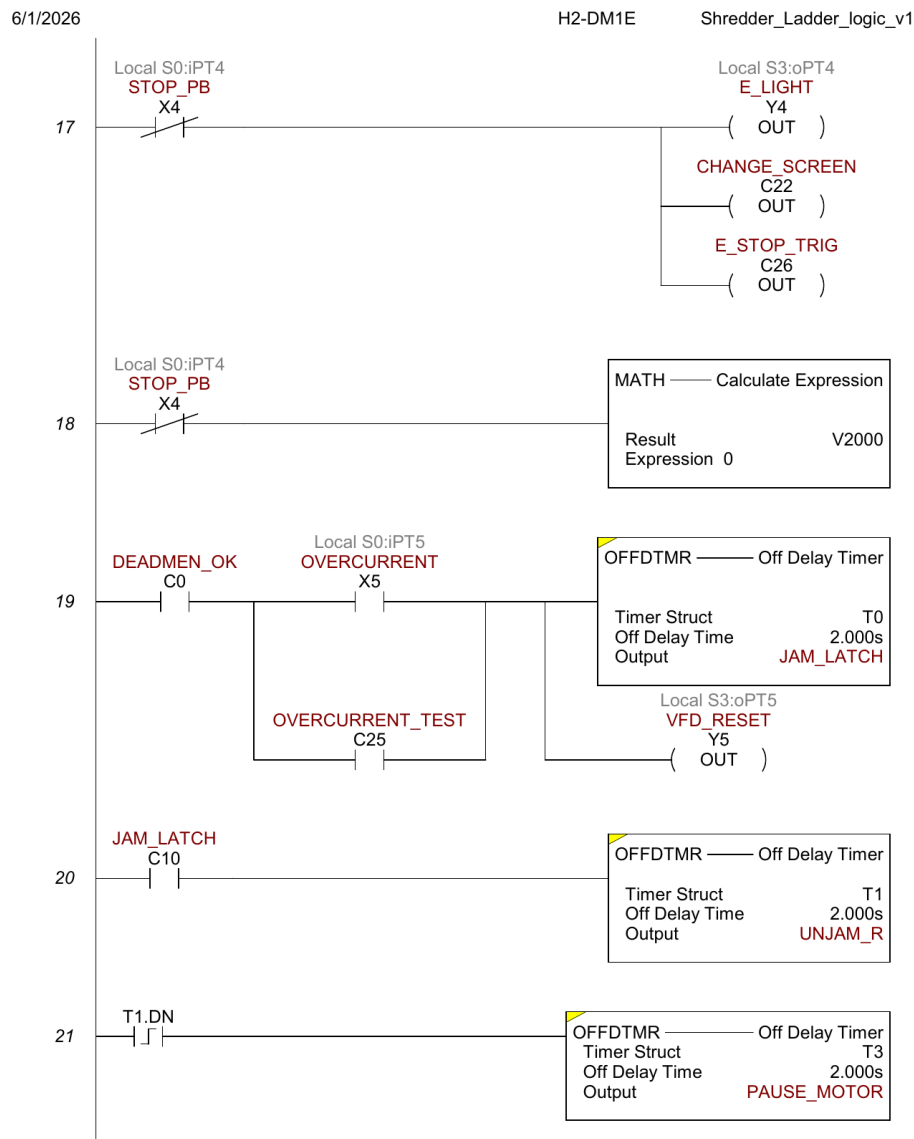


Figure 4: Ladder export page 3. Indicator LEDs and direction latches (rungs 11–16).

- **Rung 11**  $((X2 \text{ OR } C23) \text{ AND NOT } Y4 \rightarrow \text{OUT } Y2)$ : deadman A LED lights when the button is pressed but only if the fault indicator (Y4) is not lit.
- **Rung 12**  $((X3 \text{ OR } C24) \text{ AND NOT } Y4 \rightarrow \text{OUT } Y3)$ : identical pattern for deadman B.
- **Rung 13**  $(X0 \rightarrow \text{SET } C8 + \text{RST } C9)$ : forward pushbutton latches FORWARDS\_SET and clears REVERSE\_SET (mutual exclusion).
- **Rung 14**  $(X1 \rightarrow \text{SET } C9 + \text{RST } C8)$ : reverse pushbutton is the mirror image.
- **Rung 15**  $(C8 \rightarrow \text{OUT } Y0)$ : forward indicator LED ON whenever FORWARDS\_SET is latched.
- **Rung 16**  $(C9 \rightarrow \text{OUT } Y1)$ : reverse indicator LED ON whenever REVERSE\_SET is latched.

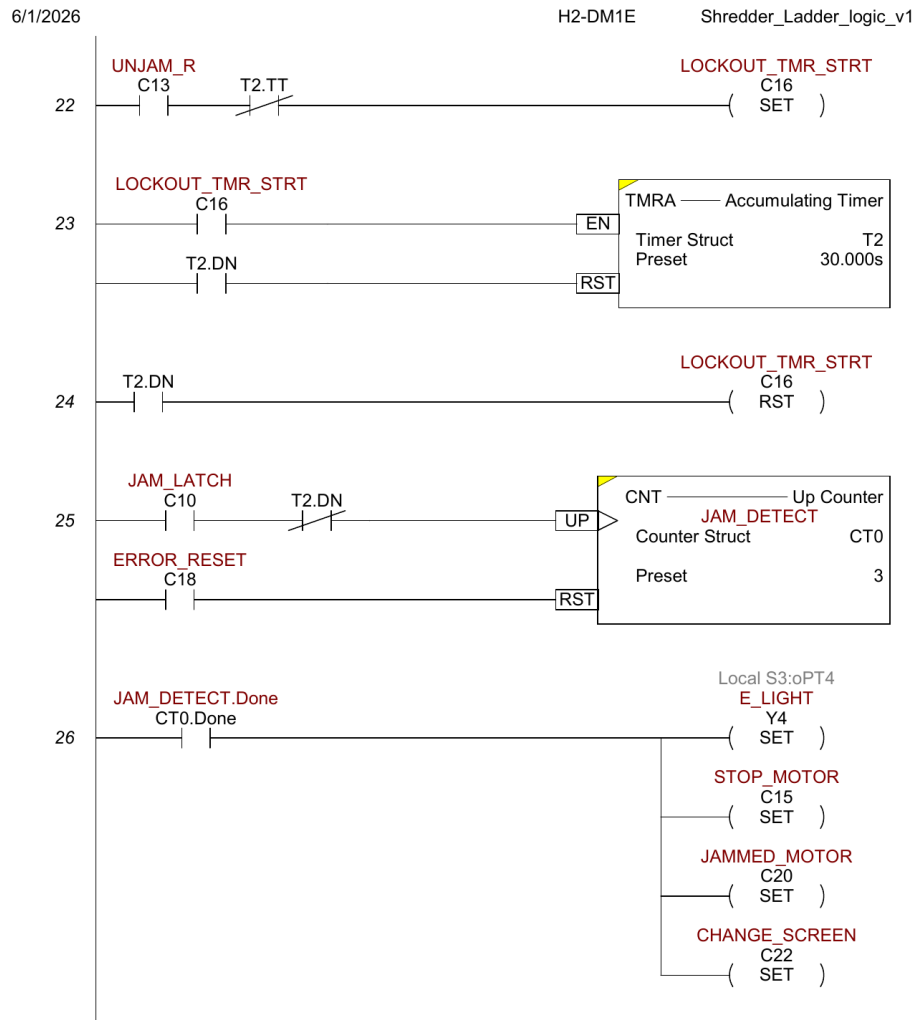
## Page 4 of 7 — Rungs 17–21: E-stop Response, Jam Detection, and Unjam Stroke



**Figure 5:** Ladder export page 4. E-stop response, jam detection, and unjam stroke (rungs 17–21).

- **Rung 17** (NOT X4 → OUT Y4 + OUT C22 + OUT C26): when E-stop opens (X4 LOW), immediately light the fault LED, set the screen-change flag for HMI Screen 3, and set the E\_STOP\_TRIG indicator.
- **Rung 18** (NOT X4 → MATH V2000=0): when E-stop opens, zero the speed setpoint to prevent a speed step on power restore.
- **Rung 19** (C0 AND (X5 OR C25) → OFFDTMR T0 (2 s) → C10 + OUT Y5): jam detection. While deadman is held and the VFD fault relay is active (or HMI soft test is active), JAM\_LATCH (C10) goes HIGH immediately and stays HIGH 2 s after the input drops; Y5 (VFD\_RESET) is energized directly to pulse the VFD RST terminal.
- **Rung 20** (C10 → OFFDTMR T1 (2 s) → C13): unjam reverse stroke timer. UNJAM\_R (C13) stays HIGH for 2 s after JAM\_LATCH drops, so rung 10 forces Y7 (reverse) for at least 2 s of recovery.
- **Rung 21** (T1.DN rising edge → OFFDTMR T3 (2 s) → C17): post-reverse pause. When UNJAM\_R finishes, T3 holds PAUSE\_MOTOR HIGH for 2 s so the motor can coast to rest before re-accelerating forward.

## Page 5 of 7 — Rungs 22–26: Accumulating Lockout Timer and Jam Counter

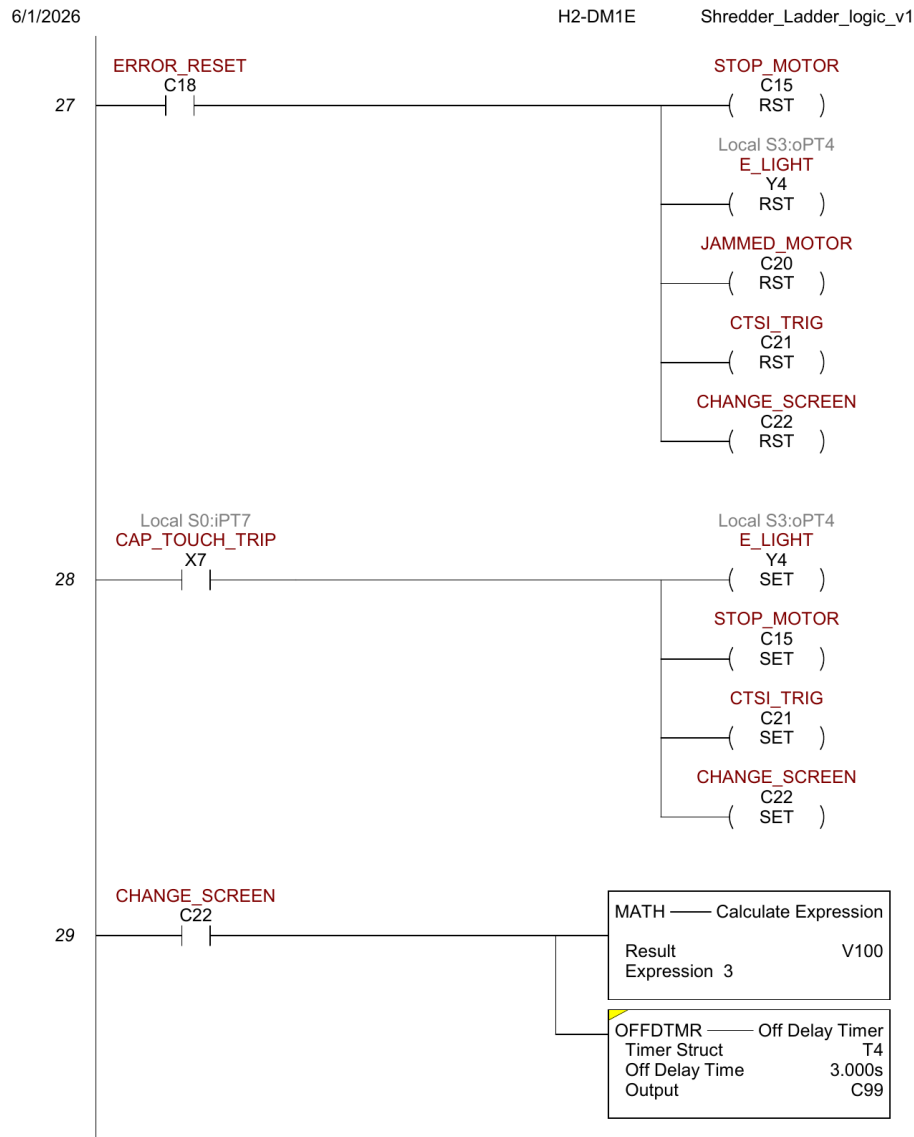


**Figure 6:** Ladder export page 5. Accumulating lockout timer and jam counter (rungs 22–26).

- **Rung 22** (C13 AND NOT T2.TT → SET C16): starts the jam-count accumulating timer. The NOT T2.TT condition prevents re-triggering if T2 is mid-count.
- **Rung 23** (C16 → TMRA T2 EN; T2.DN → RST T2): 30-second accumulating timer. T2.DN resets itself and rung 24 clears C16, stopping accumulation.
- **Rung 24** (T2.DN → RST C16): after 30 s of accumulated unjam activity, stop accumulation. New JAM\_LATCH events no longer increment CT0.
- **Rung 25** (C10 AND NOT T2.DN → CNT UP CT0 (preset 3) [RST: C18]): jam counter increments each time JAM\_LATCH goes HIGH while T2 has not completed. Preset 3 means CT0.Done fires after the third jam.
- **Rung 26** (CT0.Done → SET Y4 + SET C15 + SET C20 + SET C22): lockout on third jam. Y4 fault LED on permanently, STOP\_MOTOR hard-stops the motor (rungs 9 and 10 block), JAMMED\_MOTOR triggers the Screen 3 indicator, CHANGE\_SCREEN switches to Screen 3.



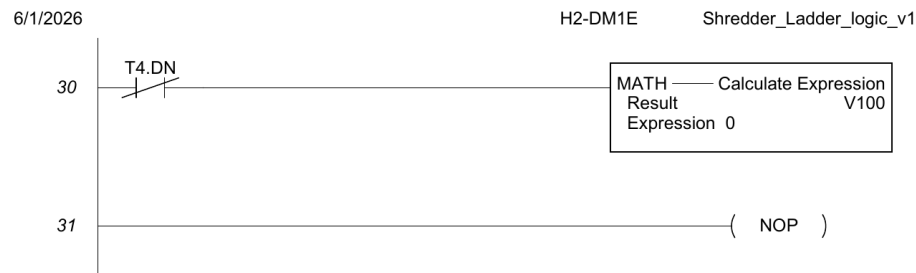
## Page 6 of 7 — Rungs 27–29: Fault Reset, Capacitive Touch, HMI Screen Change



**Figure 7:** Ladder export page 6. Fault reset path, capacitive touch trip, and HMI screen change (rungs 27–29).

- **Rung 27** ( $C18 \rightarrow RST\ C15 + RST\ Y4 + RST\ C20 + RST\ C21 + RST\ C22$ ): fault reset path (from HMI Screen 3 SHREDDER JAM or CAP TOUCH button, tag ERROR\_OFF = C18).
- **Rung 28** ( $X7 \rightarrow SET\ Y4 + SET\ C15 + SET\ C21 + SET\ C22$ ): capacitive touch safety trip. Any HIGH on X7 immediately sets all four lockout flags.
- **Rung 29** ( $C22 \rightarrow MATH\ V100=3 + OFFDTMR\ T4\ (3\ s) \rightarrow C99$ ): HMI screen navigation. MATH writes 3 to V100 (ERROR\_SCREEN tag) so the HMI switches to Screen 3.

Page 7 of 7 — Rungs 30–31: Screen Reset and End of Program



**Figure 8:** Ladder export page 7. Screen reset timeout and end-of-program NOP (rungs 30–31).

- **Rung 30** (T4.DN  $\rightarrow$  MATH V100=0): after 3 s, V100 is reset to 0, releasing the HMI from the forced screen-3 display.
- **Rung 31** (NOP): no-operation placeholder marking the end of the program.

### 6.1.5 Jam Recovery Sequence — Complete Timeline

The trace below shows exactly what happens from overcurrent detection through resume, based on the actual timer values and rung logic:

| Time                          | Event / Signal                                                         | Active outputs and coils                                                                                                         |
|-------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| $t = 0$                       | X5 (VFD fault relay) goes HIGH while C0 (DEAD-MEN_OK) is HIGH          | T0 input HIGH $\rightarrow$ C10 (JAM.LATCH) HIGH; Y5 (VFD.RESET) HIGH; C13 (UNJAM.R) HIGH; Y7 (reverse) HIGH; Y6 LOW             |
| $t = 0^+$                     | VFD fault cleared by Y5 pulse; X5 drops LOW                            | T0 off-delay starts (2 s); Y5 goes LOW; C10 stays HIGH; C13 stays HIGH; Y7 stays HIGH                                            |
| $t = 2$ s                     | T0 off-delay expires $\rightarrow$ C10 (JAM.LATCH) goes LOW            | C10 LOW $\rightarrow$ T1 off-delay starts (2 s); C13 (UNJAM.R) stays HIGH; Y7 stays HIGH; CT0 increments                         |
| $t = 4$ s                     | T1 off-delay expires $\rightarrow$ C13 (UNJAM.R) goes LOW; T1.DN rises | Y7 goes LOW (reverse stops); T1.DN rising edge triggers T3; C17 (PAUSE.MOTOR) HIGH                                               |
| $t = 6$ s                     | T3 off-delay expires $\rightarrow$ C17 (PAUSE.MOTOR) goes LOW          | Rung 9 NOT C17 condition clears; Y6 (forward) re-energizes if deadman still held                                                 |
| Lockout at $t = 0$ of 3rd jam | CT0.ACC = 3 $\rightarrow$ CT0.Done HIGH                                | C15 (STOP.MOTOR) SET; C20 (JAMMED.MOTOR) SET; Y4 SET; C22 $\rightarrow$ Screen 3. Motor locked until C18 (ERROR.RESET) from HMI. |

## 6.2 HMI Programming

The HMI is an AutomationDirect EA1-T4CL 4-inch C-more Micro-Graphic touch panel running the `HMI.413.mgp` project, connected to the H2-DM1E CPU over the Do-more serial protocol. The project has 4 screens and 27 PLC tags.

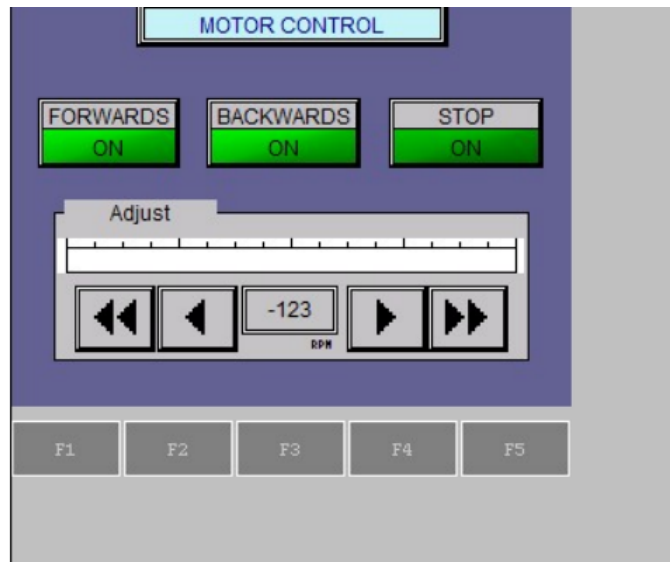
### 6.2.1 HMI Tag Database

Every PLC address visible on the HMI is documented below. Tags without a PLC address are local HMI variables (used for navigation, transient state, or labels).

| #  | Tag Name      | Data Type     | PLC Address | What it represents                                                      |
|----|---------------|---------------|-------------|-------------------------------------------------------------------------|
| 1  | AUTO          | Discrete      | —           | HMI-local; not used in current build                                    |
| 2  | CTSL_TRIG     | Discrete      | C21         | Capacitive touch safety trip flag (SET by X7 rung)                      |
| 3  | CURRENT       | Signed int 16 | V3000       | Scaled motor current (0–140 range, from WX0 via rung 3 SCALE)           |
| 4  | D-MAN A       | Discrete      | C23         | TESTMAN_A_OK — HMI soft deadman A (Test screen)                         |
| 5  | D-MAN B       | Discrete      | C24         | TESTMAN_B_OK — HMI soft deadman B (Test screen)                         |
| 6  | DEADMEN_OK    | Discrete      | C0          | Both deadman buttons (physical or test) held simultaneously             |
| 7  | ERROR_LIGHT   | Discrete      | Y4          | Error indicator output — physical panel fault LED                       |
| 8  | ERROR_OFF     | Discrete      | C18         | Error reset coil — clears all latched faults when momentarily SET       |
| 9  | ERROR_SCREEN  | BCD int 16    | V100        | HMI screen selector — written to 3 on fault, 0 after 3 s (T4)           |
| 10 | E_STOP_TRIG   | Discrete      | C26         | E-stop triggered flag — SET when X4 (STOP_PB) opens                     |
| 11 | FREQ          | Signed int 16 | V11         | VFD output frequency in 0.1 Hz units (read from VFD via serial or calc) |
| 12 | FWD           | Discrete      | C8          | FORWARDS_SET — forward direction selected (latching bit)                |
| 13 | GRAPHIC_IND 1 | Discrete      | —           | HMI-local graphic indicator; unused                                     |
| 14 | JAMMED_MOTOR  | Discrete      | C20         | Motor jammed and locked out (SET by CT0.Done rung)                      |
| 15 | JAM_COUNTER   | BCD int 16    | N0          | CT0 accumulator — jam count 0–3 (N0 = CT0.ACC in Do-more)               |

|    |              |               |       |                                                                            |
|----|--------------|---------------|-------|----------------------------------------------------------------------------|
| 16 | MOTOR_STATUS | Discrete      | C12   | RUN_REQUEST — motor run command active                                     |
| 17 | OVERCURRENT  | Discrete      | C25   | OVERCURRENT_TEST — soft overcurrent trigger (Test screen button)           |
| 18 | RPM          | Signed int 16 | V14   | Calculated or measured shaft RPM (destination register)                    |
| 19 | RPM_CTRL     | Signed int 16 | V2000 | Speed setpoint (0–1600 counts) — scaled to WY0 (0–4095) for VFD analog out |
| 20 | RVS          | Discrete      | C9    | REVERSE.SET — reverse direction selected (latching bit)                    |
| 21 | SPEED        | Signed int 16 | —     | HMI-local; unused                                                          |
| 22 | SPEED(1)     | Signed int 16 | —     | HMI-local; unused                                                          |
| 23 | STOP         | Discrete      | —     | HMI-local; unused                                                          |
| 24 | STOP_REQ     | Discrete      | C1    | HMI.STP_REQ — operator stop request from HMI STOP button                   |
| 25 | TEMP         | Signed int 16 | V15   | Temperature register (destination, RTD module — not yet calibrated)        |
| 26 | VALUE        | Signed int 32 | —     | HMI-local; unused                                                          |
| 27 | VOLTAGE      | Signed int 16 | V13   | Voltage register (destination — source TBD)                                |

### 6.2.2 Screen 1 — MOTOR CONTROL



**Figure 9:** HMI Screen 1 — Motor Control (operator screen).

The primary operator screen. Header bar reads **MOTOR CONTROL**. Contains three toggle pushbuttons and a speed adjust panel.

- **FORWARDS button** (FWD / C8, toggle): toggles FORWARDS\_SET. ON = green, OFF = dark. Sets C8, resets C9 (rung 13).
- **BACKWARDS button** (RVS / C9, toggle): toggles REVERSE\_SET. Sets C9, resets C8 (rung 14).
- **STOP button** (STOP\_REQ / C1, toggle): toggles HMI\_STP\_REQ. Rung 7 uses NOT C1 to block RUN\_REQUEST.
- **Bar graph** (RPM\_CTRL / V2000): horizontal bar 0–1600 showing current V2000 value.
- **Numeric display** (RPM\_CTRL / V2000): 3-digit integer labeled 'RPM' below.
- **IncDec  $\pm 50$  /  $\pm 250$**  (RPM\_CTRL / V2000): fine and coarse speed adjust buttons.
- **F1/F2/F3 keys**: navigation. F1 → Screen 1, F2 → Screen 2, F3 → Screen 3.

**Speed control data path:** V2000 (RPM\_CTRL, 0–1600) is scaled by rung 2 to WY0 (0–4095), which drives the analog output module F2-08DA-2 CH1 → VFD VI terminal (0–10 V frequency reference).

### 6.2.3 Screen 2 — MOTOR TELEMETRY

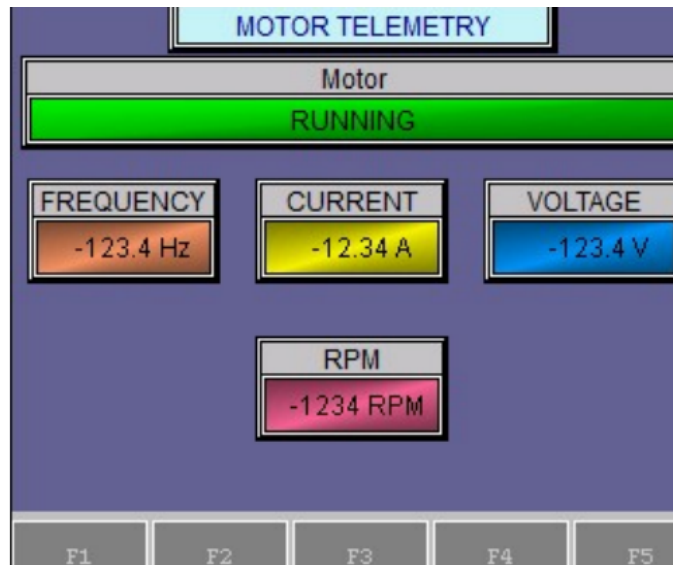


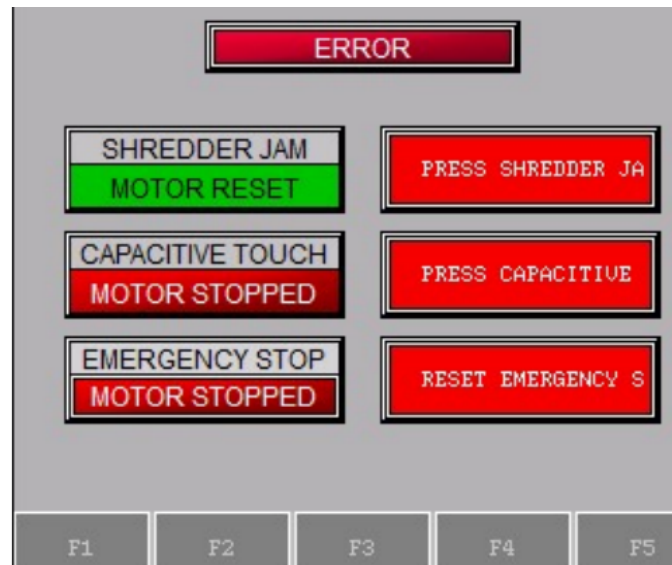
Figure 10: HMI Screen 2 — Motor Telemetry (read-only status).

Header bar reads **MOTOR TELEMETRY**. Read-only status display. A full-width indicator light spans the top and shows RUNNING (green) or STOPPED (gray).

- **Motor status indicator** (MOTOR\_STATUS / C12): full-width bar, ON = 'RUNNING' green, OFF = 'STOPPED' gray.
- **FREQUENCY display** (FREQ / V11): 4-digit, 1 decimal, Hz. Orange background.
- **CURRENT display** (CURRENT / V3000): 4-digit, 2 decimal, A. Yellow background.
- **VOLTAGE display** (VOLTAGE / V13): 4-digit, 1 decimal, V. Blue background.
- **RPM display** (RPM / V14): 4-digit, 0 decimal, RPM. Pink background.
- **F1/F2/F3 keys**: navigation. F3/F4/F10 → Screen 3.



### 6.2.4 Screen 3 — ERROR (Fault / Alarm Screen)

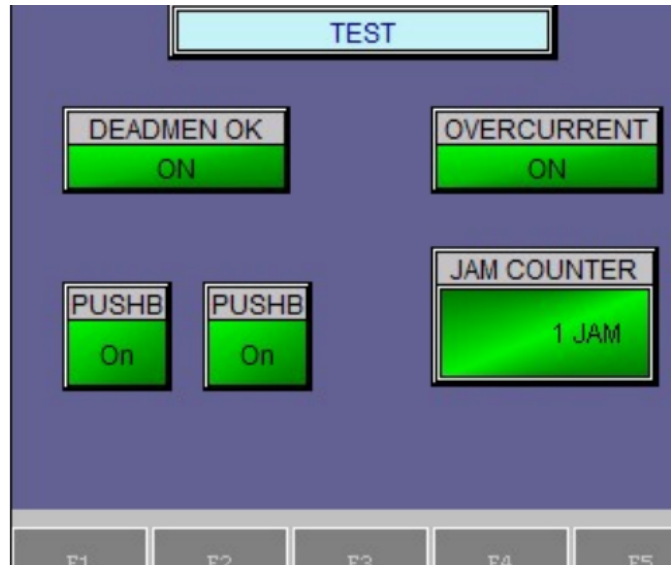


**Figure 11:** HMI Screen 3 — Error / Alarm screen (red backlight, three fault rows).

Backlight color 11 (red backlight). Automatically shown when `CHANGE_SCREEN` (C22) is SET by the ladder (rung 29 writes `V100=3`; after 3 s T4 resets `V100` to 0). Three fault rows each with a status indicator and scrolling instruction text.

- **ERROR header indicator** (`ERROR_LIGHT` / Y4): ON shows 'ERROR' in red; OFF shows 'ERROR RESET' in gray. Y4 is the physical fault LED output.
- **SHREDDER JAM row** (button + indicator): button `ERROR_OFF` / C18; indicator `JAMMED_MOTOR` / C20. Indicator lit red ('MOTOR STOPPED') when C20 HIGH (CT0.Done fired). Button pulses C18 (`ERROR_RESET`).
- **CAPACITIVE TOUCH row** (button + indicator): button `ERROR_OFF` / C18; indicator `CTSI_TRIG` / C21. Indicator lit red when C21 HIGH (capacitive touch trip).
- **EMERGENCY STOP row** (indicator only): `E_STOP_TRIG` / C26. ON shows 'MOTOR STOPPED' red; OFF shows 'MOTOR RESET' green.
- **Scroll text:** each row scrolls instructions: 'PRESS SHREDDER JAM BUTTON TO RESET MOTOR.', 'PRESS CAPACITIVE TOUCH TRIGGER BUTTON TO RESET MOTOR.', 'RESET EMERGENCY STOP BUTTON TO RESET MOTOR.'

### 6.2.5 Screen 4 — TEST (Engineering / Commissioning)



**Figure 12:** HMI Screen 4 — Test / Commissioning (soft inputs for bench testing).

Header reads **TEST**. Used during commissioning to simulate inputs without the physical pushbuttons. Not accessible from normal operation screens (only reachable via F5 from Screen 3).

- **DEADMEN OK button** (DEADMEN\_OK / C0, toggle): directly forces C0 HIGH. Allows testing run logic without holding physical buttons.
- **OVERCURRENT button** (OVERCURRENT / C25, momentary on): pulses C25 (OVERCURRENT\_TEST). Rung 19 treats C25 same as X5, triggering the full jam recovery sequence.
- **JAM COUNTER display** (JAM\_COUNTER / N0, BCD): shows CT0 accumulator as jam count 0–3.
- **PUSHBUTTON A** (D-MAN A / C23, toggle): forces C23 (TESTMAN\_A\_OK) HIGH. Rung 5 uses C23 as alternate to X2 for deadman A.
- **PUSHBUTTON B** (D-MAN B / C24, toggle): forces C24 (TESTMAN\_B\_OK) HIGH. Rung 5 uses C24 as alternate to X3 for deadman B.

## 7 Safety Design

## 7.1 NFPA 79 Compliance

- Wire color coding follows NFPA 79 §13.2: Black for AC line/load, Red for AC control, Blue for DC control, Blue/W stripe for DC return, Green/Yellow for ground.
- This color convention is applied to every conductor in the panel.
- All control wires labeled at both ends.
- Control wires land as straight stripped conductors at the terminal blocks (no ferrules used in this build).

## 7.2 Safety Functions

- **E-stop:** Category 0 stop (immediate removal of power). Hardwired — an E22B1 NC contact in series with the SD-N35 contactor coil; pressing it drops the coil and opens the contactor independently of the PLC. Single channel; the PLC additionally monitors the state on X4 (not a safety-rated channel).
- **Two-hand / deadman control:** both blue pushbuttons must be held continuously to enable the motor. As-built, the two NO inputs are AND-ed in the standard PLC's ladder logic to gate the VFD run output. It is the run-permissive and the safety interlock satisfying requirement M6, but it is *not* a safety-rated two-hand device per ISO 13851 Type III (no simultaneity timer, complementary contacts, or self-monitoring).
- **Access guarding:** handled by the ME team's mechanical design (feed geometry), so no electrical lid interlock is implemented (a lid interlock was a may-have, not a requirement).
- **Capacitive-touch module (planned):** a capacitive-touch safety module is on hand and the ladder logic to act on it (input X7 → stop motor + fault indication) is written, but it is *not yet wired*. This is a may-have (additional safety sensing) and is not relied upon in the current build.
- **Overload:** the VFD's electronic motor-overload function plus the over-torque trip (see Jam Detection); equipment protection rather than a personnel-safety function.

## 7.3 Performance Level (ISO 13849-1) — Preliminary Assessment

This is a preliminary, self-assessed determination from the as-built architecture. A formal verification needs a documented ISO 12100 risk assessment (to set the required performance level, PLr) and component reliability data (MTTFd / B10d, diagnostic coverage). Table 2 summarizes the achieved categories.

**Table 2:** Preliminary ISO 13849-1 assessment of the as-built safety functions.

| Safety function        | As-built architecture                                                                                                                                                           | Category (prelim.)                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Emergency stop         | Hardwired E22B1 NC contact in series with the SD-N35 contactor coil; opening it removes motor power. Single channel, well-tried components; PLC monitors X4 (not safety-rated). | Cat 1 (PL b–c)                       |
| Two-hand / deadman     | Two NO buttons AND-ed in the standard PLC to gate the VFD run output. Single channel; no simultaneity check, complementary contacts, or self-monitoring.                        | Cat B (PL b); not ISO 13851 Type III |
| Over-torque / jam stop | VFD over-torque trip plus electronic motor overload. Equipment protection, not a personnel-safety function.                                                                     | Not rated (process protection)       |

Overall the control system is approximately **Category B–1 (PL b–c)**, limited by: single-channel architecture throughout; a standard (non-safety) PLC evaluating the two-hand permissive; and a single contactor with no force-guided (mechanically linked) contacts and no feedback (EDM) monitoring to detect a welded contactor. The hardwired E-stop, built from well-tried components, is the strongest function (Category 1).

To raise the achieved PL — for example to Category 3 / PL d, should the risk assessment require it for the rotating-blade hazard — would need a dual-channel architecture, a safety relay or safety-rated PLC, two contactors (or force-guided contacts) with feedback monitoring, and a synchronous two-hand module meeting ISO 13851.

## 7.4 Failure Mode Analysis

- Wire break in jam signal: PLC sees X5 stuck LOW. Jam signal cannot fire, but the hardwired E-stop loop and motor contactor provide independent safety. Operator can also see motor stop and trigger investigation via fault-light absence.
- VFD loses power: the FA–FC fault relay de-energizes and opens, so X5 reads LOW (indistinguishable from healthy). The motor stops because the VFD itself is unpowered, so the condition is caught by the motor not running rather than by the jam signal.
- PLC fails: motor contactor drops out (safety chain bypasses PLC). Motor coasts to stop.
- Brake resistor disconnected: VFD will trip on overvoltage during deceleration. Motor coasts to stop. Diagnosable from VFD display.

## 8 User Manual

### 8.1 Purpose of This Manual

This manual provides operators, maintenance personnel, and safety officers with the procedures necessary to safely operate, monitor, and perform basic troubleshooting on the Universal Plastic Shredder V2.0 electrical control system. It assumes the user has read the full engineering report and understands the basic hazards described in Section 7 (Safety Design).

### 8.2 Intended Audience and Required Training

- **Operators:** Must complete a site-specific safety briefing covering two-hand deadman operation, E-stop use, and jam recovery. No electrical troubleshooting is permitted without additional training.
- **Maintenance personnel:** Must be familiar with NFPA 70E arc-flash boundaries, lock-out/tagout (LOTO) per 29 CFR 1910.147, and the contents of Appendix B (I/O point list) and Appendix D (VFD parameters).
- **Supervisors / safety officers:** Shall maintain a copy of this manual and the full engineering report adjacent to the machine or in a controlled electronic repository.

### 8.3 Normal Operating Procedure

#### 8.3.1 Pre-Start Checks

1. Verify the main disconnect (PowerTec 71008) is in the OFF position.
2. Inspect the feed area for obstructions, loose plastic, or foreign objects.
3. Verify the E-stop button is pulled out (released).
4. Ensure both deadman pushbuttons are not pressed.
5. Close and latch any mechanical access guards (ME team's responsibility).

#### 8.3.2 Startup

1. Turn the main disconnect to ON. The HMI shall display **IDLE / READY**. The green status indicator shall illuminate.
2. If the HMI shows a fault or lockout message, press the HMI **RESET** button (see Section 8.6).
3. To start shredding:

- Press and **hold both blue deadman pushbuttons simultaneously**.
  - The motor will accelerate to the programmed speed (70–90 RPM at the shaft).
  - The red “RUNNING” status indicator shall illuminate.
  - If either deadman button is released, the motor stops immediately.
4. Feed plastic sheet material at a moderate, steady rate. Avoid overstuffing.

### 8.3.3 Normal Shutdown

1. Release both deadman buttons. The motor will coast to a stop.
2. Wait for the blades to come to a complete stop (minimum 5 seconds).
3. Turn the main disconnect to OFF.

## 8.4 Emergency Stop and Safety Functions

- **To stop in an emergency:** Press the red mushroom E-stop button. Motor power is removed immediately (Category 0 stop). The HMI will show **ESTOP ACTIVE**.
- **To reset after E-stop:** Twist the E-stop button clockwise to release it, then press the HMI RESET button.
- **Two-hand deadman:** The motor runs only while both blue buttons are held. This is a permissive — not a safety-rated two-hand control per ISO 13851 Type III. Never bypass or defeat this function.

## 8.5 Jam Detection and Automatic Recovery

The control system includes an auto-reverse jam recovery feature. When an over-torque jam is detected (motor current exceeds 150% of rated for 3 seconds):

1. The VFD trips and the motor stops.
2. The HMI displays **JAM DETECTED** and increments the jam counter.
3. The PLC automatically:
  - (a) Sends a 500 ms reset pulse to clear the VFD fault.
  - (b) Runs the motor in reverse for 2 seconds to clear the jam.
  - (c) Pauses briefly, then restarts forward operation.
4. If the jam clears, normal operation resumes.

5. If a jam occurs again within the same operating session:

- Jam counter increments.
- After the **third jam** within a session, the system enters **LOCKOUT** mode.
- The HMI shows **LOCKOUT — MANUAL RESET REQUIRED**.
- Motor cannot restart until a supervisor presses the HMI RESET button and clears the feed area.

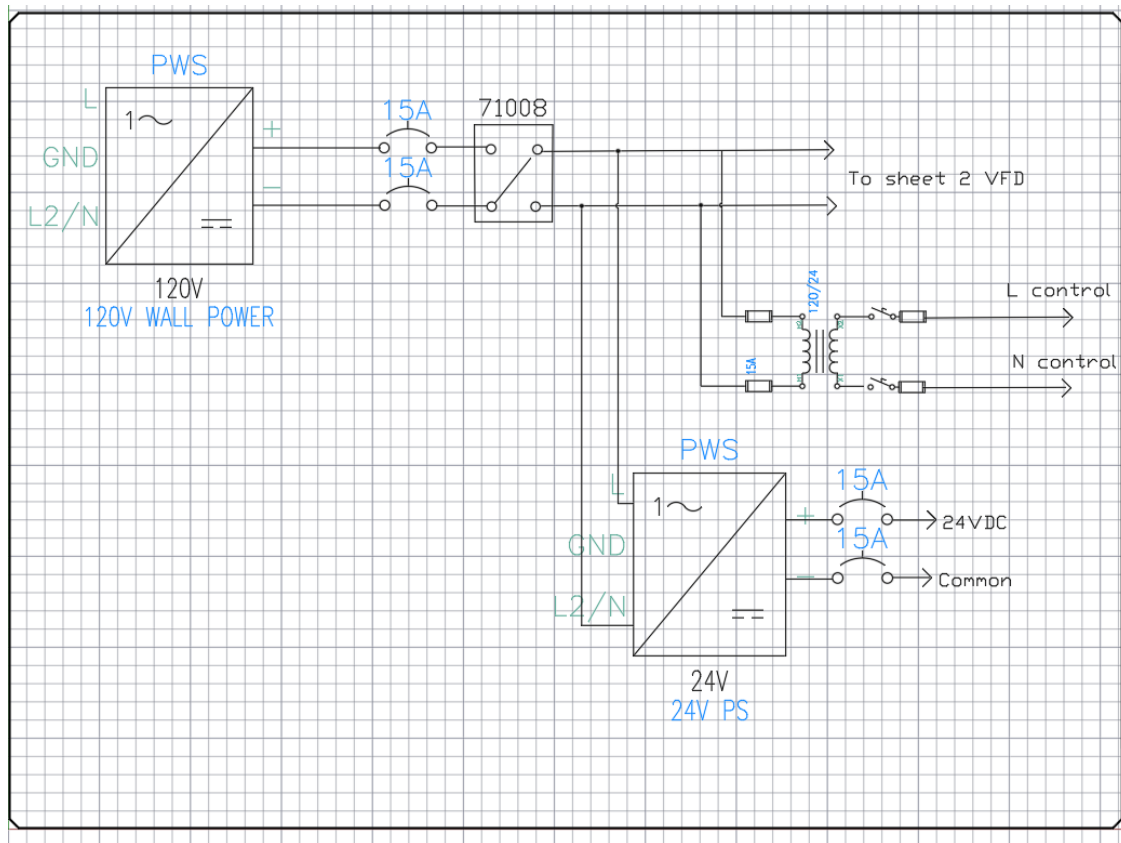
## 8.6 Fault Recovery and Lockout Reset

**Table 3:** Fault codes and recovery actions

| HMI Display      | Operator Action                                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------------------------------|
| ESTOP ACTIVE     | Release E-stop button, then press HMI RESET.                                                                          |
| VFD FAULT        | Check VFD display for code (overvoltage, overtemp, etc.). Power cycle main disconnect. If repeated, call maintenance. |
| JAM (1 or 2)     | No action — automatic recovery runs. Monitor HMI.                                                                     |
| LOCKOUT (3 jams) | Clear plastic from blades manually (after LOTO), then press HMI RESET.                                                |
| IDLE / READY     | Normal.                                                                                                               |

## 8.7 Schematic and Wiring Diagrams

The following figures show the core electrical circuits of the shredder control system. All wire colors follow NFPA 79 §13.2 (see Section 7.1). Full terminal-level details are in Appendix B (I/O point list).

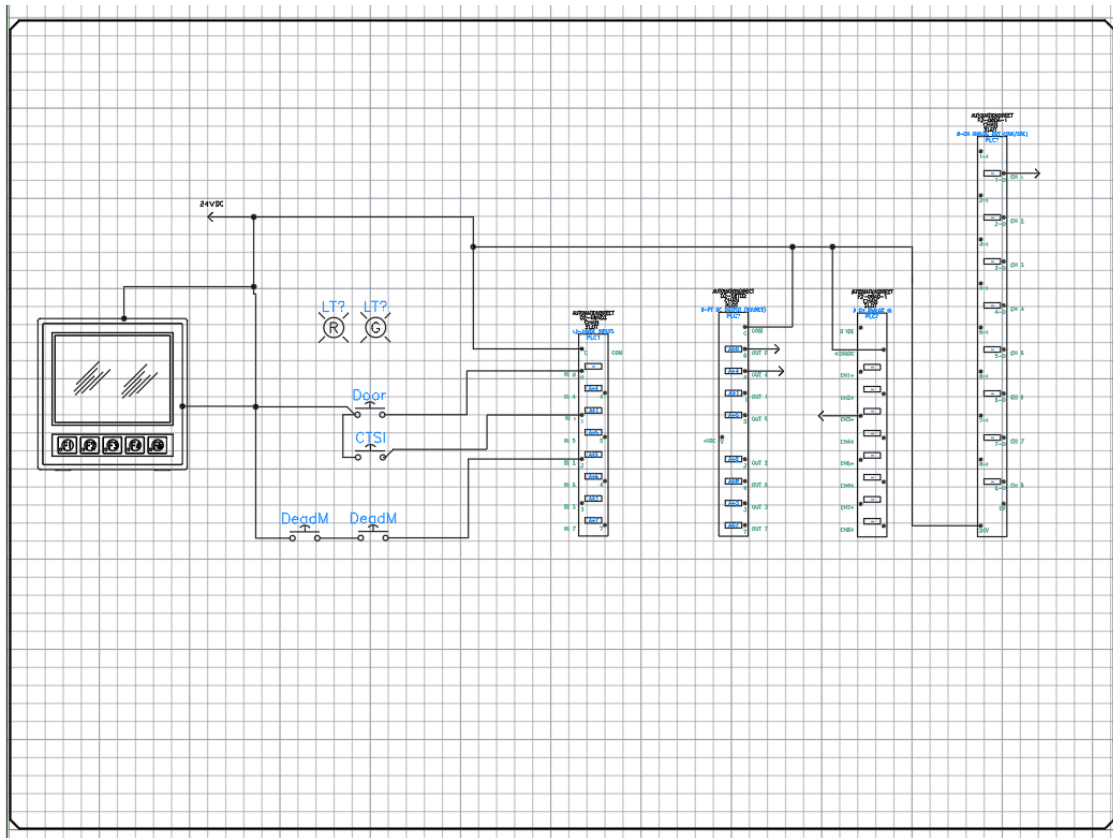


**Figure 13:** Power distribution circuit: main disconnect, branch breakers, 24 V power supply, and VFD input.





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**Figure 15:** PLC I/O circuit: deadman buttons, E-stop monitor, VFD fault input, and VFD run/rev/reset outputs.

## 8.8 Daily and Periodic Maintenance

### 8.8.1 Operator Daily Checks

- Before first use: Verify E-stop button stops the motor immediately.
- Check that both deadman buttons are responsive (motor runs only when both held).
- Listen for unusual noises (grinding, excessive vibration).
- Ensure the HMI jam counter resets to zero after a clean 30-second run.

### 8.8.2 Monthly Maintenance (Qualified Personnel Only)

- Perform LOTO (lockout/tagout) on the main disconnect.
- Open the electrical enclosure.
- Visually inspect for loose wires, discoloration, or burned terminals.
- Check that the brake resistor (75 ohm, 500w) shows no cracking or overheating signs.

- Verify fan operation on the VFD and power supply.
- Clean dust from the enclosure using low-pressure compressed air.
- Close and secure enclosure before re-energizing.

## 8.9 Troubleshooting Guide

**Table 4:** Common issues and corrective actions

| Symptom                            | Possible cause                | Action                                                     |
|------------------------------------|-------------------------------|------------------------------------------------------------|
| Motor does not start               | E-stop pressed                | Release E-stop, reset HMI.                                 |
|                                    | Only one deadman pressed      | Press both simultaneously.                                 |
|                                    | LOCKOUT active                | Clear jam, press HMI RESET.                                |
| Motor runs briefly then stops      | Jam detected                  | Let auto-reverse run. If repeated, clear plastic manually. |
| HMI blank                          | No 24V power                  | Check Mean Well PSU input breaker (10A) and output fuses.  |
| VFD shows fault code               | Overvoltage / overtemp / etc. | See VFD manual Appendix D. Power cycle.                    |
| Jam counter increments without jam | Over-torque threshold too low | Check PD124. Nominal = 150%.                               |

## 8.10 Safety Warnings Summary

### CRITICAL SAFETY WARNINGS

- Never reach into the feed area while blades are moving. Wait for complete stop.
- Do not bypass, tape down, or modify the deadman buttons or E-stop.
- Lockout/tagout the main disconnect before any enclosure opening or blade service.
- This machine has no electrical feed interlock — the mechanical feed geometry is the primary access safeguard.
- If the automatic jam recovery cycles three times, **stop and manually clear** the jam.

## 8.11 Contact for Service and Support

For maintenance beyond this manual, contact the Ichor Systems facilities team or the original design team (via Portland State University, ECE Department). Have the following ready:

- Shredder serial number (TBD — assign upon final assembly)
- VFD fault code (if any)
- HMI jam count and lockout status
- Photos of any visible damage

## 8.12 Revision History of This Manual

| Version | Date       | Changes                        |
|---------|------------|--------------------------------|
| 1.0     | 2026-05-31 | Initial release (rough draft). |

# 9 Standards and Regulatory Compliance

This section consolidates every electrical and machine-safety standard the design is built against and states, for each, how it is applied to the shredder. Many of these are referenced in passing elsewhere in this report; they are gathered here so a reviewer (or a future maintenance team) can audit compliance in one place.

## 9.1 Standards Summary Matrix

| Standard      | Scope                                                      | How it is applied here                                                                                 |
|---------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| NFPA 70 (NEC) | National Electrical Code — premises wiring, motor circuits | Branch-circuit, conductor, grounding, and disconnect design (see §9.2).                                |
| NFPA 79       | Electrical standard for industrial machinery               | Wire color coding, supply disconnect, protective bonding, stop functions, operator devices (see §9.3). |
| NEMA 250      | Enclosure type ratings                                     | Control enclosure selected to a NEMA type matched to the lab/shop environment (see §9.4).              |

*continued on next page...*

| Standard             | Scope                                        | How it is applied here                                                          |
|----------------------|----------------------------------------------|---------------------------------------------------------------------------------|
| NEMA ICS 1/2         | Industrial control & systems                 | Contactor / control device selection and marking.                               |
| NEMA MG 1            | Motors and generators                        | Motor nameplate interpretation and VFD parameterization.                        |
| UL 508A              | Industrial control panels                    | Panel construction practice and short-circuit current rating (SCCR) awareness.  |
| IEC 60204-1          | Safety of machinery – electrical equipment   | International counterpart to NFPA 79; used to cross-check stop categories.      |
| ISO 12100            | Risk assessment / risk reduction             | Framework behind the hazard analysis driving the safety circuit.                |
| ISO 13849-1          | Safety-related parts of control systems (PL) | Preliminary self-assessment (Cat B–1) of the E-stop and two-hand control chain. |
| ISO 13850            | Emergency stop function                      | E-stop actuator type, color, and Category 0 behavior.                           |
| ISO 13851            | Two-hand control devices                     | Deadman two-hand start timing and configuration.                                |
| OSHA 29 CFR 1910.147 | Lockout/Tagout (LOTO)                        | Lockable main disconnect for energy isolation during service.                   |

## 9.2 NFPA 70 – National Electrical Code

| NEC article               | Requirement                         | How it is applied here                                                                                                                     |
|---------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Art. 430.6                | Motor FLC basis                     | From the GE test-motor nameplate (FLA 7.6 A at 230 V); basis for all downstream sizing. Re-verified when the production motor is selected. |
| Art. 430.22               | Branch conductor sizing             | Conductors sized at $\geq 125\%$ of motor FLC.                                                                                             |
| Art. 430.32               | Running overload                    | VFD electronic motor-overload (PD142 = 7.6 A) plus the over-torque trip.                                                                   |
| Art. 430.52               | Branch short-circuit / ground-fault | 30 A C-curve breaker on the VFD branch; clears inrush without nuisance trips (inverse-time-breaker allowance).                             |
| Art. 430.102/.109         | Disconnecting means                 | Provided ahead of the controller and VFD.                                                                                                  |
| Art. 430 Part X (430.122) | Adjustable-speed drive systems      | Conductor and protection rules applied to the VFD-fed motor circuit.                                                                       |
| Art. 250                  | Grounding & bonding                 | EGC (green / green-yellow) bonds the VFD, PSU, motor frame, and panel to a common ground bus and building earth.                           |
| Art. 310                  | Conductor ampacity                  | Conductors chosen from the ampacity tables for the installed temperature rating.                                                           |
| Art. 409                  | Industrial control panels           | Panel layout, marking, and overcurrent considerations.                                                                                     |
| Art. 110.26               | Working space                       | Clearances around live parts maintained for safe service; final panel mounting confirmed to leave the required clearance.                  |

### 9.3 NFPA 79 – Electrical Standard for Industrial Machinery

| NFPA clause | 79 Requirement               | How it is applied here                                                                                                                                            |
|-------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| §5.3        | Supply disconnecting means   | Lockable main disconnect (PowerTec 71008) isolates the machine from supply; also satisfies LOTO.                                                                  |
| §7          | Overcurrent protection       | Branch and downstream fusing at terminal blocks (PSU 25 A slow-blow, PLC 1 A fast-blow, control loads individually fused).                                        |
| §8          | Protective bonding circuit   | Continuous equipment-grounding / bonding path to all exposed conductive parts.                                                                                    |
| §9          | Control circuits & functions | Stop functions classified by stop category; the hardwired safety chain has precedence over the PLC.                                                               |
| §10.7–10.8  | Operator interface / E-stop  | Red mushroom actuator on a yellow background, self-latching NC contact; start devices guarded by the two-hand requirement.                                        |
| §13.2       | Conductor color coding       | Black AC line/load, red AC control, blue DC control, blue/white DC return, green/yellow PE; wires labeled both ends, landed straight (no ferrules in this build). |

### 9.4 NEMA Standards

| Standard           | Covers                       | How it is applied here                                                                                                  |
|--------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| NEMA 250           | Enclosure type ratings       | PLC/HMI control box is sealed for Type 12 (dust-tight, drip-tight) duty in the lab/shop environment.                    |
| NEMA ICS 1 / ICS 2 | Industrial control & systems | Contactors and control-device selection/markings; the Mitsubishi SD-N35 is sized above motor FLA for AC-3 (motor) duty. |
| NEMA MG 1          | Motors & generators          | Interpreting the motor nameplate (voltage, poles, RPM, service factor) for VFD parameters PD141--PD144.                 |

## 9.5 Machine-Safety Standards (ISO / IEC)

| Standard    | Covers                            | How it is applied here                                                                                                            |
|-------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| ISO 12100   | Risk assessment / reduction       | Identifies the primary hazards (rotating blades, entanglement, electrical) and drives the layered safeguards.                     |
| ISO 13849-1 | Safety-related control parts (PL) | Preliminary assessment: hardwired E-stop $\approx$ Cat 1; two-hand permissive $\approx$ Cat B (see the Safety Functions section). |
| ISO 13850   | Emergency stop function           | Category 0 stop (immediate removal of power) with fail-safe NC wiring.                                                            |
| ISO 13851   | Two-hand control devices          | Both buttons actuated within 0.5 s of each other and held continuously to enable the motor.                                       |
| IEC 60204-1 | Electrical equipment of machines  | International analogue of NFPA 79; used to cross-check stop categories and supply disconnection.                                  |

## 9.6 Occupational Safety (OSHA)

| Regulation            | Covers                | How it is applied here                                                                                       |
|-----------------------|-----------------------|--------------------------------------------------------------------------------------------------------------|
| 29 CFR 1910.147       | Lockout/Tagout (LOTO) | The main supply disconnect is lockable for positive isolation during blade-area service and maintenance.     |
| 29 CFR 1910 Subpart S | Electrical safety     | General electrical-safety practices (guarding of live parts, grounding) observed throughout the panel build. |

**[TODO: Before final submission: convert any remaining “targets / intends to” language above into verified compliance statements and attach the panel enclosure datasheet.]**

# 10 Testing and Validation

## 10.1 Signal Chain Tests

- Manual jumper test (FA-FC short): confirmed PLC X5 reads LOW correctly – wiring intact.
- Guaranteed-trip test (PD124 = 1, PD125 = 5.0): VFD trips reliably on motor run, fault relay fires, PLC X5 drops.
- Threshold tuning: 150% (PD124 = 150) chosen for production based on no-load current (0.5 A) versus expected jam current (above 11 A actual).



## 10.2 Load Testing

[TODO: Schedule load testing with actual plastic samples once mechanical integration complete. Need ME team coordination.]

## 10.3 Safety Validation

- E-stop response time: [TODO: measure with scope].
- Two-hand control timing: [TODO: verify both buttons must be held continuously; simultaneous release stops within X ms].

# 11 Results

## 11.1 What Works

- All motor parameters programmed and verified.
- Over-torque detection chain validated (VFD detects, relay fires, PLC reads, ladder reacts).
- Fail-safe NC wiring confirmed – system correctly enters fault state on signal chain failure.
- HMI communicates with the PLC over the serial COM link.

## 11.2 Open Issues

- Vendor-specific Huanyang firmware quirk: PD052 = 12 does not fire relay in continue-running mode. Working around with PD052 = 02 + PD123 = 3.
- PLC ladder logic not fully complete – state machine implemented, but unjam routine timing needs tuning with real loaded motor.
- Mechanical integration pending – have not yet tested under real shredding load.
- Motor-current monitor (VFD V0 → F2-08AD-1 CH1) is wired, but under time constraints it was never configured/scaled correctly to read the VFD output current. The signal path exists; the VFD analog-output and PLC analog-input scaling still need to be set and calibrated.

[QUESTION: How much detail should we include about the Huanyang firmware quirk? It is an interesting design lesson (vendor lock-in, never trust the datasheet completely) but distracts from the high-level results.]

## 12 Discussion

### 12.1 Design Tradeoffs

- Cost versus reliability: chose Huanyang VFD (\$150) over ATO (\$450). Firmware quirks documented in this report (the Software Design note and Appendix D) for future maintenance.
- 110V single-phase versus 220V split-phase mains: 110V chosen for receptacle availability in PSU lab; trade-off is double the input current and need for a dedicated 30 A circuit.
- NO active-high versus NC fail-safe jam signal: the VFD fault relay is wired NO active-high (FA–FC closes on trip, X5 HIGH). Simpler to wire, but not fail-safe at the signal level—a broken wire reads as “no fault.” That residual risk is covered by the independent hardwired E-stop loop and motor contactor, which remove power regardless of the PLC.

### 12.2 Lessons Learned

- Always verify VFD relay function with multimeter before assuming firmware works as documented.
- Document deviations from datasheet behavior in version control so future teams can find them.
- Power cycle after parameter changes – some VFDs require it for save to fully take effect.
- Use ferrules on control wires in future builds: this build landed straight stranded wire at the terminal blocks, and loose/spread strands caused multiple debugging hours during commissioning.

### 12.3 Project Management Notes

The team ran on a weekly cadence with three standing touchpoints. Internal working meetings were held once a week to plan tasks, hand off blocking items between members, and review the running TODO list. Faculty review with Dr. Jonathan Bird (EE advisor) ran on either Thursday or Tuesday each week depending on lab and class conflicts; these sessions focused on technical direction (VFD parameterization, jam recovery design, safety architecture) and caught issues like the over-torque relay firmware quirk and the two-hand permissive scope decision before they propagated into the build. Industry coordination with Bridget Lannigan at MME / Ichor Systems happened on Wednesdays and was the primary forum for ME/EE integration, requirements clarifications, and confirming the as-built scope against the sponsor’s needs. Prof. Andrew Greenberg attended the broader capstone review checkpoints across both terms.

All source artifacts (this report, the Do-more ladder `.dmd` project and its `Circuit_Logic.pdf` export, the C-more Micro HMI `HMI_413.mgp` project, the AutomationDirect Point List workbook, the BOM workbook, the AutoCAD Electrical project, the wiring documentation, and every supporting markdown reference) live in a single GitHub repository. Tracking the full edit history in version control gave the team a shared audit trail and a working backup, and it is what makes the rung-by-rung walkthrough in Section 6.1 reproducible: every figure in this report points back to a committed source file.

Overall, the schedule held to the rough-draft milestone. VFD selection and parameterization landed in Winter term; panel build, point-list finalization, and the ladder logic finished in early Spring; HMI programming and the safety integration filled the back half of Spring. Items that slipped (final production motor selection, capacitive-touch wiring as a may-have, and inline insertion of the exported AutoCAD wiring diagrams) are tracked as `\todo` markers throughout this document so they stay visible until closed.

## 13 Future Work and Improvements

### 13.1 Safety and Compliance

- **Integrate the capacitive-touch safety (CTSI) module:** wire the on-hand capacitive-touch module to input X7. The ladder logic to stop the motor and raise a fault on a touch trip is already written, so this is primarily a wiring and validation task. It adds a no-contact operator-presence safeguard at the feed.
- **Raise the safety performance level:** move from the current single-channel Cat B-1 toward Cat 3 / PL d — a safety relay or safety-rated PLC, dual-channel inputs, two contactors (or force-guided contacts) with feedback (EDM) monitoring, and a synchronous two-hand module per ISO 13851.
- **Ferrule the control wiring:** crimp ferrules on stranded control conductors at the terminal blocks to eliminate the loose-strand faults seen during commissioning.

### 13.2 Monitoring and Connectivity

- **Telemetry:** stream live operating data (motor current, speed, jam count, fault state, runtime) off the PLC for dashboards and trend analysis.
- **Modbus to the VFD:** the Huanyang supports RS-485 Modbus (currently unused). Adding it would pull far richer data straight from the drive — output current, voltage, frequency/RPM, DC-bus voltage, drive temperature, and detailed fault codes — beyond the single 0–10 V analog monitor.

- **Internet / remote viewing:** expose a read-only view of device status over the network — via the H2-DM1E’s built-in web server or an ESP32 / edge gateway — so the shredder can be monitored remotely from a browser or phone.
- **Data logging:** record total runtime, jam events per shift, and faults locally for predictive maintenance.

### 13.3 Capability

- **Predictive jam detection:** train a small model on the motor-current waveform shape just before a jam to detect and pre-empt jams earlier.
- **Mechanical integration:** add a feed hopper that auto-meters plastic into the blades.
- **Higher-capacity model:** scale to a 5–10 HP drivetrain for greater throughput.

## 14 Ethics and Design Considerations

A summary of the ethics and design evaluation follows:

- **Public health and safety:** Yes – layered safety functions (hardwired E-stop, two-hand deadman control), NFPA 70 / NEMA compliance, hardwired safety chain.
- **Environmental:** Yes – core project purpose is diverting plastic from landfill.
- **Economic:** Yes – significant cost savings versus commercial alternatives, careful component selection within budget.
- **Global:** Possibly – approach is replicable globally, components internationally sourced.
- **Cultural:** No – industrial machine for B2B use, no culture-specific design.

## 15 References

### Codes and Standards

- NFPA 70 (2023). *National Electrical Code* (Articles 110, 250, 310, 409, 430). National Fire Protection Association, Quincy, MA.  
[nfpa.org/codes-and-standards](https://www.nfpa.org/codes-and-standards) (NFPA 70)
- NFPA 79 (2024). *Electrical Standard for Industrial Machinery*. NFPA, Quincy, MA.  
[nfpa.org/codes-and-standards](https://www.nfpa.org/codes-and-standards) (NFPA 79)

- NEMA 250 (2020). *Enclosures for Electrical Equipment (1000 Volts Maximum)*. National Electrical Manufacturers Association.  
[nema.org](http://nema.org) (NEMA 250)
- NEMA ICS 1 (2000, R2020). *Industrial Control and Systems: General Requirements*.  
[nema.org](http://nema.org) (NEMA ICS 1)
- NEMA ICS 2 (2000, R2020). *Industrial Control and Systems: Controllers, Contactors, and Overload Relays Rated 600 Volts*.  
[nema.org](http://nema.org) (NEMA ICS 2)
- NEMA MG 1 (2021). *Motors and Generators*.  
[nema.org](http://nema.org) (NEMA MG 1)
- UL 508A (2018). *Standard for Industrial Control Panels*. Underwriters Laboratories.  
[shopulstandards.com](http://shopulstandards.com) (UL 508A)
- IEC 60204-1:2016. *Safety of Machinery: Electrical Equipment of Machines, Part 1*. International Electrotechnical Commission.  
[webstore.iec.ch/publication/26037](http://webstore.iec.ch/publication/26037)
- ISO 12100:2010. *Safety of Machinery: General Principles for Design, Risk Assessment and Risk Reduction*. International Organization for Standardization.  
[iso.org/standard/51528.html](http://iso.org/standard/51528.html)
- ISO 13849-1:2023. *Safety of Machinery: Safety-related Parts of Control Systems, Part 1*. ISO.  
[iso.org/standard/73481.html](http://iso.org/standard/73481.html)
- ISO 13850:2015. *Safety of Machinery: Emergency Stop Function*. ISO.  
[iso.org/standard/59970.html](http://iso.org/standard/59970.html)
- ISO 13851:2019. *Safety of Machinery: Two-Hand Control Devices*. ISO.  
[iso.org/standard/70295.html](http://iso.org/standard/70295.html)
- OSHA 29 CFR 1910.147. *The Control of Hazardous Energy (Lockout/Tagout)*. U.S. Department of Labor, Occupational Safety and Health Administration.  
[osha.gov/laws-regs/regulations/standardnumber/1910/1910.147](http://osha.gov/laws-regs/regulations/standardnumber/1910/1910.147)

## Manufacturer Datasheets and Manuals

- Huanyang Electrical, *HY-series Variable Frequency Drive User Manual* (HY02D211B-T).  
[huanyanginverter.com](http://huanyanginverter.com)
- AutomationDirect, *DL205 User Manual* (D2-USER-M).  
[automationdirect.com](http://automationdirect.com) (D2-USER-M)

- AutomationDirect, *Do-more H2 Series CPU User Manual* (HX-USER-M).  
automationdirect.com (HX-USER-M)
- AutomationDirect, *C-more Micro-Graphic Panels User Manual* (EA-MG-USER-M).  
automationdirect.com (EA-MG-USER-M)
- Mean Well, *NDR-480 Series 480 W Single Output Industrial DIN Rail Power Supply Datasheet*.  
meanwell.com (NDR-480)
- Mitsubishi Electric, *SD-N Series Magnetic Contactors Catalog* (SD-N35).  
mitsubishielectric.com (SD-N magnetic contactors)
- ATO, *APCS-300R30-AD 300 W 30  $\Omega$  Brake Resistor Datasheet*.  
ato.com (300 W 30  $\Omega$  braking resistor)
- General Electric, *5KE Frame 182T Motor Nameplate / Service Data* (5KE182BC205B).  
Archived service literature.
- PowerTec, *71008 Two-Pole START/STOP Switch Product Page*.  
powertecmotors.com
- Cutler-Hammer (Eaton), *E22 Series 22 mm Pushbutton Operator Catalog* (E22B1 mushroom E-stop).  
eaton.com (E22 series)

## Course and Sponsor Materials

- ECE 412/413 Senior Capstone Project Development I/II course materials and project proposal, Portland State University, Maseeh College of Engineering and Computer Science, Winter and Spring 2026.
- MME / Ichor Systems, *Universal Plastic Shredder V2.0 Sponsor Brief*, Bridget Lannigan, 2026 (internal communications, archived in repository).
- PSU Power Engineering Laboratory, *EE 347L Switchgear Project Final Report Template* (basis for the document structure and PSU MCECS report style used here).

## 16 Appendices

### 16.1 Appendix A: Bill of Materials

The complete as-built Bill of Materials is consolidated in `BOM/BOM_Combined_As_Built.xlsx` (and the rendered `BOM/BOM_Combined_As_Built.pdf`) in the project repository. The workbook contains three sheets:

- **Combined BOM** — every item actually installed in the control panel, grouped by category (Motor & Drive, Power Distribution, PLC Control System, HMI, Operator Controls, Motor Connections, Wire/Duct/Hardware) with manufacturer, part number, quantity, and notes pointing back to the relevant documentation.
- **Excluded - Not Used** — items that appeared in preliminary BOMs but were not used in the final build, with the reason (rejected during selection, not purchased, superseded by another component, or out of project budget).
- **VFD Decision Matrix** — the weighted comparison used to select the Huanyang HY02D211B-T (price 50%, current capability 20%, build quality 15%, documentation 10%, brand 5%).

**Critical items** (highlighted in the workbook): Motor (GE 5KE182BC205B for bench test), VFD (Huanyang HY02D211B-T), Brake Resistor (ATO APCS-300R30-AD), Mean Well NDR-480-24 24V DC power supply, Mitsubishi SD-N35 contactor, Cutler-Hammer E22B1 E-stop, AutomationDirect PLC rack with H2-DM1E CPU and the D2-08ND3 / F2-08AD-1 / F2-04RTD / D2-08TR / F2-08DA-2 modules, EA1-T4CL HMI.

*[NOTE: The bench-test motor is the GE 5KE182BC205B (3 HP, 230 V, 7.6 A, 4-pole, 1750 RPM) used to validate the control system. The final production motor remains TBD per the 3-5 HP project requirement; its nameplate will supersede VFD parameters PD141–PD144. Control-panel detail (rack layout, terminal-by-terminal wiring) is itemized in the point list in Appendix B.]*

The rendered workbook follows on the landscape pages below.

|                         |            |                          |
|-------------------------|------------|--------------------------|
| <b>PRODUCT NAME</b>     | Shredder   | <b>EE TEAM</b>           |
| <b>APPROVED BY</b>      |            | <b>Fearghus Tyler</b>    |
| <b>DATE OF APPROVAL</b> | 2026-06-01 | <b>Bao Nguyen</b>        |
| <b>PART COUNT</b>       |            | <b>61 Yaqoub Rabiah</b>  |
| <b>TOTAL COST</b>       |            | <b>\$969.88 Fox Kang</b> |

| PART NUMBER            | PART NAME                  | DESCRIPTION                                                                                                                                                                                                                                                                                                             | QUANTITY | UNITS | LINK                                                                                                                                                                                                                                                                                                        | UNIT COST | AMOUNT   |
|------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|
| 5KE182BC205B           | AC Motor (3 HP)            | GE 3-phase TEFC induction motor; 230 V (low-voltage), 1750 RPM, 7.6 A FLA, NEMA 182T frame. Bench-test motor.                                                                                                                                                                                                           | 1        | EA    | <a href="https://www.electricmotorsforless.com/">https://www.electricmotorsforless.com/</a>                                                                                                                                                                                                                 | \$0.00    | \$0.00   |
| HY02D211B-T            | VFD                        | Huanyang variable frequency drive, 110 V single-phase input, 220 V 3-phase output, 2.2 kW (3 HP). Selected via decision matrix.                                                                                                                                                                                         | 1        | EA    | <a href="https://www.amazon.com/Huanyang-1-Phase-Variable-Frequency-Output-3hp/dp/B0DMF3GB38">https://www.amazon.com/Huanyang-1-Phase-Variable-Frequency-Output-3hp/dp/B0DMF3GB38</a>                                                                                                                       | \$200.00  | \$200.00 |
| APCS-300R30-AD         | Brake Resistor             | LS Electric 300 W, 30 ohm aluminum-clad braking resistor for VFD dynamic braking on P+/PR. Purchased from AutomationDirect. Note: an earlier identical-part-number unit (ATO branded) was ordered but never arrived after shipping; replaced with this LS Electric unit which is the brake resistor actually installed. | 1        | EA    | <a href="https://www.automationdirect.com/ad/shopping/catalog/motion_control/ls_electric_l7c_l7ca_series/servo_systems/servo_accessories/apcs-300r30-ad">https://www.automationdirect.com/ad/shopping/catalog/motion_control/ls_electric_l7c_l7ca_series/servo_systems/servo_accessories/apcs-300r30-ad</a> | \$29.00   | \$29.00  |
| 71008                  | Main Power Switch          | Powertec 120 V ON/OFF main disconnect switch for entire control panel.                                                                                                                                                                                                                                                  | 1        | EA    | <a href="https://powertecproducts.com/71008/">https://powertecproducts.com/71008/</a>                                                                                                                                                                                                                       | \$15.00   | \$15.00  |
| TMC 8 3C 15A (2907618) | VFD Branch Breaker         | Phoenix Contact thermal-magnetic DIN-rail circuit breaker, 15 A, on VFD neutral leg (per as-built). Obtained at no cost from the PSU EPL free pile.                                                                                                                                                                     | 1        | EA    | <a href="https://www.phoenixcontact.com/">https://www.phoenixcontact.com/</a>                                                                                                                                                                                                                               | \$0.00    | \$0.00   |
| NDR-480-24             | 24 VDC Power Supply        | Mean Well 480 W, 24 VDC DIN-rail power supply for PLC, HMI, indicator LEDs, and VFD control loops. Donated by sponsor / obtained at no cost to the EE team.                                                                                                                                                             | 1        | EA    | <a href="https://www.meanwell.com/">https://www.meanwell.com/</a>                                                                                                                                                                                                                                           | \$0.00    | \$0.00   |
| SD-N35                 | Main Power Contactor       | Mitsubishi 3-pole magnetic contactor. Energized through E-stop chain via E22B1 NC contact.                                                                                                                                                                                                                              | 1        | EA    | <a href="https://www.ebay.com/itm/...">https://www.ebay.com/itm/...</a> (tpmachinery seller, order 23-14145-39817, Jan 2026)                                                                                                                                                                                | \$12.59   | \$12.59  |
| EURO S4LH              | Fuse Block (button common) | Morsettitalia fuse-holder terminal block. Holds 1 A fast-blow fuse protecting the shared pushbutton +24 V supply.                                                                                                                                                                                                       | 1        | EA    | <a href="https://www.morsettitalia.com/">https://www.morsettitalia.com/</a>                                                                                                                                                                                                                                 | \$8.00    | \$8.00   |
| EURO S10H-5H           | Fuse Block (AC branches)   | Morsettitalia fuse-holder terminal block for AC branch protection.                                                                                                                                                                                                                                                      | 1        | EA    | <a href="https://www.morsettitalia.com/">https://www.morsettitalia.com/</a>                                                                                                                                                                                                                                 | \$12.00   | \$12.00  |
| H2-DM1E                | PLC CPU / Ethernet         | AutomationDirect Do-more CPU with built-in Ethernet, fits D2-09B-1 DL205 rack. Purchased Jan 2026 from planet-surplus as a USED combo for \$350 total, includes: D2-09B-1 chassis, D2-08ND3, F2-08AD-1, F2-04RTD, D2-08TR, F2-08DA-2, and D2-FILL filler panels.                                                        | 1        | EA    | <a href="https://www.ebay.com/itm/...">https://www.ebay.com/itm/...</a> (planet-surplus seller, order 20-14113-87209, Jan 2026)                                                                                                                                                                             | \$350.00  | \$350.00 |
| D2-09B-1               | PLC Base / Rack            | AutomationDirect DirectLOGIC 205 9-slot AC-powered base. Included in the H2-DM1E used combo purchase.                                                                                                                                                                                                                   | 1        | EA    | <a href="https://www.automationdirect.com/">https://www.automationdirect.com/</a>                                                                                                                                                                                                                           | \$0.00    | \$0.00   |
| D2-08ND3               | Discrete Input Module      | AutomationDirect 8-ch, 24 VDC, sourcing/sinking discrete input module. Slot 2. Inputs X0-X7. Included in the H2-DM1E used combo purchase.                                                                                                                                                                               | 1        | EA    | <a href="https://www.automationdirect.com/">https://www.automationdirect.com/</a>                                                                                                                                                                                                                           | \$0.00    | \$0.00   |
| F2-08AD-1              | Analog Input Module        | AutomationDirect 8-ch, 4-20 mA analog current input module. Slot 3. CH1+ reads VFD VO current monitor. Included in the H2-DM1E used combo purchase.                                                                                                                                                                     | 1        | EA    | <a href="https://www.automationdirect.com/">https://www.automationdirect.com/</a>                                                                                                                                                                                                                           | \$0.00    | \$0.00   |
| F2-04RTD               | RTD Input Module           | AutomationDirect 4-ch RTD input module. Slot 4. Installed but unused (spare). Included in the H2-DM1E used combo purchase.                                                                                                                                                                                              | 1        | EA    | <a href="https://www.automationdirect.com/">https://www.automationdirect.com/</a>                                                                                                                                                                                                                           | \$0.00    | \$0.00   |
| D2-08TR                | Relay Output Module        | AutomationDirect 8-ch relay output module. Slot 5. Drives button LEDs (Y0-Y4) and VFD FOR/REV/RST (Y6/Y7/Y5). Included in the H2-DM1E used combo purchase (eBay listing showed D2-08TD2 but seller substituted D2-08TR which is what we use).                                                                           | 1        | EA    | <a href="https://www.automationdirect.com/">https://www.automationdirect.com/</a>                                                                                                                                                                                                                           | \$0.00    | \$0.00   |
| F2-08DA-2              | Analog Output Module       | AutomationDirect 8-ch, 0-10 V analog voltage output module. Slot 6. +V1 drives VFD VI speed reference. Included in the H2-DM1E used combo purchase (eBay listing showed F2-02DA but actual unit is F2-08DA-2).                                                                                                          | 1        | EA    | <a href="https://www.automationdirect.com/">https://www.automationdirect.com/</a>                                                                                                                                                                                                                           | \$0.00    | \$0.00   |



| PART NUMBER              | PART NAME                             | DESCRIPTION                                                                                                                                                                                                            | QUANTITY | UNITS | LINK                                                                                                                                                                                      | UNIT COST | AMOUNT   |
|--------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|
| D2-FILL                  | Rack Filler Panels                    | AutomationDirect blank fill panels for unused rack slots 7-9. Included in the H2-DM1E used combo purchase.                                                                                                             | 3        | EA    | <a href="https://www.automationdirect.com/">https://www.automationdirect.com/</a>                                                                                                         | \$0.00    | \$0.00   |
| EA1-T4CL                 | HMI Touch Panel                       | AutomationDirect C-More Micro-Graphic 4-inch color touch panel HMI. Purchased used from eBay for \$148.95 (Apr 2026). Purchase included the serial cable for HMI-to-CPU communication (RS-232 to H2-DM1E serial port). | 1        | EA    | <a href="https://www.ebay.com/itm/306662618920">https://www.ebay.com/itm/306662618920</a>                                                                                                 | \$148.95  | \$148.95 |
| S2PR-P3G                 | Forward Pushbutton Head               | Autonics 22 mm illuminated green pushbutton head (Forward command - > X0).                                                                                                                                             | 1        | EA    | <a href="https://www.autonics.com/us/model/S2PR-P3GAD">https://www.autonics.com/us/model/S2PR-P3GAD</a>                                                                                   | \$1.90    | \$1.90   |
| S2PR-P3Y                 | Reverse Pushbutton Head               | Autonics 22 mm illuminated yellow pushbutton head (Reverse command - > X1).                                                                                                                                            | 1        | EA    | <a href="https://www.autonics.com/us/model/S2PR-P3YAD">https://www.autonics.com/us/model/S2PR-P3YAD</a>                                                                                   | \$1.90    | \$1.90   |
| S2PR-P3B                 | Run/Deadman Pushbutton Head           | Autonics 22 mm illuminated blue pushbutton head (Run 1 -> X2, Run 2 -> X3).                                                                                                                                            | 2        | EA    | <a href="https://www.autonics.com/us/model/S2PR-P3BAD">https://www.autonics.com/us/model/S2PR-P3BAD</a>                                                                                   | \$1.90    | \$3.80   |
| S2PR-P3R                 | Reset Pushbutton Head                 | Autonics 22 mm illuminated red pushbutton head (planned Reset -> X6).                                                                                                                                                  | 1        | EA    | <a href="https://www.autonics.com/us/model/S2PR-P3RAD">https://www.autonics.com/us/model/S2PR-P3RAD</a>                                                                                   | \$4.50    | \$4.50   |
| SA-CA                    | NO Contact Block (1NO)                | Autonics finger-safe 1NO contact block. 110 VAC/10 A. Used with each S2PR pushbutton.                                                                                                                                  | 5        | EA    | <a href="https://www.autonics.com/us/model/SA-CA">https://www.autonics.com/us/model/SA-CA</a>                                                                                             | \$2.80    | \$14.00  |
| SA-LDG                   | LED Block - Green (24 VDC)            | Autonics modular LED indicator block, green, 12-24 VDC. Forward button illumination -> Y0.                                                                                                                             | 1        | EA    | <a href="https://www.autonics.com/us/model/SA-LDG">https://www.autonics.com/us/model/SA-LDG</a>                                                                                           | \$4.50    | \$4.50   |
| SA-LDY                   | LED Block - Yellow (24 VDC)           | Autonics modular LED indicator block, yellow, 12-24 VDC. Reverse button illumination -> Y1.                                                                                                                            | 1        | EA    | <a href="https://www.autonics.com/us/model/SA-LDY">https://www.autonics.com/us/model/SA-LDY</a>                                                                                           | \$4.50    | \$4.50   |
| SA-LDB                   | LED Block - Blue (24 VDC)             | Autonics modular LED indicator block, blue, 12-24 VDC. Run 1 and Run 2 illumination -> Y2, Y3.                                                                                                                         | 2        | EA    | <a href="https://www.autonics.com/us/model/SA-LDB">https://www.autonics.com/us/model/SA-LDB</a>                                                                                           | \$4.50    | \$9.00   |
| SA-LDR                   | LED Block - Red (24 VDC)              | Autonics modular LED indicator block, red, 12-24 VDC. Reset button illumination (planned).                                                                                                                             | 1        | EA    | <a href="https://www.autonics.com/us/model/SA-LDR">https://www.autonics.com/us/model/SA-LDR</a>                                                                                           | \$4.50    | \$4.50   |
| E22B1                    | Emergency Stop Pushbutton             | Cutler-Hammer / Eaton 22 mm mushroom E-stop, NC contact. In series with SD-N35 contactor coil.                                                                                                                         | 1        | EA    | <a href="https://www.eaton.com/">https://www.eaton.com/</a>                                                                                                                               | \$25.00   | \$25.00  |
| Wamco 525                | Fault Indicator Light                 | Wamco panel-mount LED indicator, 28 VDC, 0.6 W. Driven by PLC Y4 as machine-fault indicator.                                                                                                                           | 1        | EA    | <a href="https://www.wamco.com/">https://www.wamco.com/</a>                                                                                                                               | \$15.00   | \$15.00  |
| Banana Socket (Blue)     | Banana Socket Blue                    | Generic 4 mm panel-mount banana socket, blue. Quick-disconnect for VFD U output to motor T1. Obtained at no cost from a free pile.                                                                                     | 1        | EA    |                                                                                                                                                                                           | \$0.00    | \$0.00   |
| Banana Socket (White)    | Banana Socket White                   | Generic 4 mm panel-mount banana socket, white. VFD V output to motor T2. Obtained at no cost from a free pile.                                                                                                         | 1        | EA    |                                                                                                                                                                                           | \$0.00    | \$0.00   |
| Banana Socket (Red)      | Banana Socket Red                     | Generic 4 mm panel-mount banana socket, red. VFD W output to motor T3. Obtained at no cost from a free pile.                                                                                                           | 1        | EA    |                                                                                                                                                                                           | \$0.00    | \$0.00   |
| G1X1LG6                  | Slotted Wire Duct, 1"x1"              | Panduit Type G wide-finger slotted PVC duct, light gray, 6 ft lengths.                                                                                                                                                 | 6        | FT    | <a href="https://www.panduit.com/">https://www.panduit.com/</a>                                                                                                                           | \$3.43    | \$20.58  |
| C1LG6                    | Duct Cover, 1"W                       | Panduit duct cover for G1X1LG6, light gray, 6 ft lengths.                                                                                                                                                              | 6        | FT    | <a href="https://www.panduit.com/">https://www.panduit.com/</a>                                                                                                                           | \$1.20    | \$7.20   |
| 18 AWG Blue              | Control Wire, Blue                    | 18 AWG stranded blue control wire (NFPA 79 §13.2 DC control). Obtained at no cost (some from team member's existing stock, some from free pile).                                                                       | 1        | Roll  |                                                                                                                                                                                           | \$0.00    | \$0.00   |
| 18 AWG White             | Control Return Wire                   | 18 AWG stranded white wire for DC return (NFPA 79 §13.2). Obtained at no cost (some from team member's existing stock, some from free pile).                                                                           | 1        | Roll  |                                                                                                                                                                                           | \$0.00    | \$0.00   |
| 12 AWG Black             | Power Wire, Black                     | 12 AWG black wire for AC mains and motor leads. Obtained at no cost (some from team member's existing stock, some from free pile).                                                                                     | 1        | Roll  |                                                                                                                                                                                           | \$0.00    | \$0.00   |
| 12 AWG Green/Yellow      | Ground Wire                           | 12 AWG green/yellow stranded equipment grounding wire (NFPA 79). Obtained at no cost (some from team member's existing stock, some from free pile).                                                                    | 1        | Roll  |                                                                                                                                                                                           | \$0.00    | \$0.00   |
| GMA-500mA or 0217.500MXP | Fuse - 500 mA F                       | Bussmann or Littelfuse 5x20 mm fast-blow fuse, 500 mA, 250 V. (Currently not installed per as-built note - retained as recommended addition.)                                                                          | 1        | EA    |                                                                                                                                                                                           | \$1.50    | \$1.50   |
| GMA-1A or equiv.         | Fuse - 1 A F                          | 5x20 mm fast-blow fuse, 1 A, 250 V. Protects shared +24 V pushbutton common (in EURO S4LH).                                                                                                                            | 1        | EA    |                                                                                                                                                                                           | \$1.50    | \$1.50   |
| LM2596                   | DC-DC Buck Converter (24V to 5V/3.3V) | Seloky LM2596-based adjustable DC-DC buck converter module with voltmeter display (sold as 2-pack). Steps down 24 VDC panel supply to the input voltage required by the capacitive touch module.                       | 1        | Pack  | <a href="https://www.amazon.com/Converter-Adjustable-Regulator-Voltmeter-Display/dp/B0F1M2SR31">https://www.amazon.com/Converter-Adjustable-Regulator-Voltmeter-Display/dp/B0F1M2SR31</a> | \$7.99    | \$7.99   |

| PART NUMBER | PART NAME               | DESCRIPTION                                                                                                                                                           | QUANTITY | UNITS | LINK                                                          | UNIT COST | AMOUNT   |
|-------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|---------------------------------------------------------------|-----------|----------|
| SEB001      | Control Panel Enclosure | Vorksmen industrial control enclosure (steel) housing the entire electrical system including PLC rack, VFD, contactor, PSU, terminal blocks, and operator panel.      | 1        | EA    | <a href="https://www.amazon.com/">https://www.amazon.com/</a> | \$59.98   | \$59.98  |
| AE14921a    | 3V Relay                | AEDIKO 3 VDC PCB-mount relay. Used in the capacitive touch module sub-circuit to interface the low-voltage touch sensor output with the 24 VDC panel control voltage. | 1        | EA    | <a href="https://www.amazon.com/">https://www.amazon.com/</a> | \$6.99    | \$6.99   |
| TOTAL PARTS |                         |                                                                                                                                                                       | 61       |       |                                                               | TOTAL     | \$969.88 |

| PART NUMBER            | PART NAME                         | DESCRIPTION                                                                      | FROM BOM FILE                                    | REASON NOT USED                                                                                                         |
|------------------------|-----------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| PE145TC-2-4            | Motor (alternate)                 | Original BOM candidate motor                                                     | BOM_02_20_26                                     | Not used; GE 5KE182BC205B used instead for bench testing                                                                |
| NBE7                   | Circuit Breaker                   | Original BOM candidate breaker                                                   | BOM_1_29 / BOM_02_20_26                          | Never purchased; replaced by Phoenix Contact TMC 8 3C 15A                                                               |
| (30 A fuse, TBD)       | 30 A Branch Fuse                  | Per old BOM, 2x 30 A fuses budgeted                                              | BOM_1_29                                         | Never needed in final design                                                                                            |
| 1866-1441-ND           | DC Power Supply                   | 120/24 VDC PSU from Digi-Key                                                     | BOM_1_29                                         | Superseded by Mean Well NDR-480-24                                                                                      |
| (generic)              | VEVOR 3HP 120V VFD                | Hobby-grade VFD option                                                           | VFD_120V_3HP_Decision_Matrix                     | Rejected during selection (hobby-grade, limited documentation)                                                          |
| (generic)              | ATO 3HP 120V VFD                  | Higher-quality VFD option                                                        | VFD_120V_3HP_Decision_Matrix                     | Too expensive (\$450-500) for project budget                                                                            |
| (generic)              | 110V 20A 3HP VFD                  | Mid-tier VFD option                                                              | VFD_120V_3HP_Decision_Matrix                     | Huanyang HY02D211B-T selected instead                                                                                   |
| (WEG single-phase 3HP) | WEG Single Phase Motor            | 1-phase general purpose 3 HP motor                                               | Motor Spec                                       | Considered but GE 5KE182BC205B used                                                                                     |
| NT3-18-213T            | Worldwide Electric Motor          | 3-phase TEFC 3 HP motor                                                          | Motor Spec                                       | Considered but not purchased                                                                                            |
| (VEVOR 5HP 230V)       | VEVOR High Torque Motor           | 5 HP 1-phase compressor-style motor                                              | Motor Spec                                       | Considered but not purchased                                                                                            |
| L510 (2HP / 3HP)       | TECO L510 VFD                     | VFD comparison option                                                            | Motor Spec                                       | Considered but Huanyang selected                                                                                        |
| G0532                  | Marathon Gearmotor                | Integrated gearbox motor                                                         | Motor Spec                                       | Considered but not purchased                                                                                            |
| KBWC-15K               | KB Electronics speed control      | Triac speed control fan-style                                                    | Motor Spec                                       | VFD approach used instead                                                                                               |
| TTP223B                | Capacitive Touch Sensor           | Single-channel cap touch                                                         | BOM_1_30_beta (safety draft)                     | Does not meet industrial standards                                                                                      |
| CAP1293-1-SN           | Capacitive Touch Chip             | 3-channel cap touch chip                                                         | BOM_1_30_beta (safety draft)                     | Chip only; needs interface board                                                                                        |
| 1864-1032-ND           | Capacitive Proximity Sensor       | Cap proximity sensor                                                             | BOM_1_30_beta (safety draft)                     | Too expensive (\$109.25)                                                                                                |
| F3S-TGR-NMPC-21-10     | Omron Magnetic Safety Sensor      | Non-contact magnetic safety switch                                               | BOM_1_30_beta (safety draft)                     | Too expensive; out of stock                                                                                             |
| RSS 36-I2-D-R-ST5      | Schmersal RFID Safety Sensor      | RFID safety sensor                                                               | BOM_1_30_beta (safety draft)                     | Suspected industrial-only / non-retail                                                                                  |
| F3SG-SR / PG           | Omron Safety Light Curtain        | Safety light curtain                                                             | BOM_1_30_beta (safety draft)                     | Out of scope for capstone budget                                                                                        |
| (Ichor-provided)       | Red/Green Indicator Lights        | Sponsor-provided lights                                                          | BOM_02_20_26                                     | Not used; Wamco 525 fault light and button LEDs cover indication                                                        |
| (Ichor-provided)       | E-stop Pushbutton                 | Sponsor-provided E-stop                                                          | BOM_02_20_26                                     | Replaced with Cutler-Hammer/Eaton E22B1 procured by team                                                                |
| V30AE000055            | Insulated Ferrules, 12 AWG (gray) | Z+F crimp ferrules, 12 AWG, 10 mm length, gray (DIN color code). 500-piece pack. | URS Electronics invoice (purchased but not used) | Purchased from URS Electronics but ultimately not used in the as-built panel. Spare stock retained for future projects. |

## 16.2 Appendix B: PLC I/O Point List

The complete as-built point list, transcribed from the project workbook (`Point_List.xlsx`), follows on the landscape pages below — every sheet, using the spreadsheet’s columns *Point Description*, *Origin Address*, *DO / DI / AO / AI / Pwr*, *Destination Address*, *Destination Description*, and *Notes*: the index; the Mitsubishi SD-N35 contactor; the Mean Well NDR-480-24 power supply; the DirectLOGIC 205 rack and the H2-DM1E CPU; the EA1-T4CL HMI; every I/O module (D2-08ND3, F2-08AD-1, F2-04RTD, D2-08TR, F2-08DA-2), the legacy D2-08TA and F2-02DAS-2 sheets; and the Huanyang HY02D211B-T VFD.

## Sheet: Index

| POINT<br>INDEX | ASSIGNMENT |               |                            |  |  |
|----------------|------------|---------------|----------------------------|--|--|
|                | PAGE NO.   | NO. OF SHEETS | DESCRIPTION                |  |  |
|                | 1          | 1             | Mitsubishi SD-N35          |  |  |
|                | 2          | 2             | Mean Well NDR-480-24       |  |  |
|                | 3          | 3             | Direct Logic 205 D2-09B-1  |  |  |
|                | 4          | 4             | DirectLOGIC CPU (H2-DM1E)  |  |  |
|                | 5          | 5             | Output Module (D2-08TA)    |  |  |
|                | 6          | 6             | HMI (EA1-T4CL)             |  |  |
|                | 7          | 7             | Output Module (F2-02DAS-2) |  |  |
|                | 8          | 8             | Input Module (D2-08ND3)    |  |  |
|                | 9          | 9             | Analog Input (F2-08AD-1)   |  |  |
|                | 10         | 10            | RTD Input (F2-04RTD)       |  |  |
|                | 11         | 11            | Relay Output (D2-08TR)     |  |  |
|                | 12         | 12            | Analog Output (F2-08DA-2)  |  |  |
|                | 13         | 13            | VFD (Huanyang HY02D211B-T) |  |  |

Sheet: Index — Revision History

|  | REV | DATE       | DESCRIPTION                                                 | Created By | Checked by     |
|--|-----|------------|-------------------------------------------------------------|------------|----------------|
|  | 1   | 2026-03-14 | Initial point list (end of Winter term).                    | Bao Nguyen | Fearghus Tyler |
|  | 2   | 2026-05-18 | Updated with as-built wiring and module corrections.        | Bao Nguyen | Fearghus Tyler |
|  | 3   | 2026-06-01 | Added missing component sheets to index; full revision log. | Bao Nguyen | Fearghus Tyler |

## Sheet: Mitsubishi SD-N35

| Point                                       | Description  | Origin | DO | DI | AO | AI | Pwr | Destination Address | Destination Description               | Notes                                                                |
|---------------------------------------------|--------------|--------|----|----|----|----|-----|---------------------|---------------------------------------|----------------------------------------------------------------------|
| Contactor (Mitsubishi SD-N35) – Coil A1     |              | A1     | 0  | 0  | 0  | 0  | 1   | Y0                  | D2-08TD2 Output Module (12–24VDC)     | A1 → DF103V (1/2A slow blow fuse) → D2-08TD2 output supplying +24VDC |
| Contactor (Mitsubishi SD-N35) – Coil A2     |              | A2     | 0  | 0  | 0  | 0  | 1   | -V                  | NDR-480-24 Power Supply               | A2 → direct return to power supply negative (–V)                     |
| Contactor (Mitsubishi SD-N35) – Line Input  |              | L1     | 0  | 0  | 0  | 0  | 1   | L                   | Phoenix Contact TMC 83C 15A (2907618) | L1 → Phoenix Contact fuse breaker → fed from wall AC line            |
| Contactor (Mitsubishi SD-N35) – Load Output |              | T1     | 0  | 0  | 0  | 0  | 1   | R                   | HY Series VFD (HY02D211B-T)           | T1 → passes through contactor → supplies VFD terminal R (line input) |
|                                             | Total Points | 0      | 0  | 0  | 0  | 0  | 4   |                     |                                       |                                                                      |

## Sheet: Mean Well NDR-480-24

| Point                                                    | Descrip-<br>tion | Origin | DO | DI | AO | AI | Pwr | Destination<br>Address                                     | Destination<br>Descrip-<br>tion                                                                                                            | Notes                                                                                                                                                                                 |
|----------------------------------------------------------|------------------|--------|----|----|----|----|-----|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                          |                  |        | 0  |    |    |    |     | AC Mains Hot from wall via main power switch and 25 A fuse | 120 VAC line conductor from wall outlet, routed through panel main disconnect, then through 25 A slow-blow fuse on EURO S10H-5H fuse block | Wall AC L -> main disconnect -> EURO S10H-5H 25 A slow-blow -> PSU L. Black wire per NFPA 79.                                                                                         |
| Mean Well NDR-480-24 N (AC Neutral Input)                |                  | N      | 0  | 0  | 0  | 0  | 1   |                                                            | 120 VAC neutral conductor from wall outlet, returns through neutral bus                                                                    | Wall AC N -> neutral bus -> PSU N. White wire per NFPA 79.                                                                                                                            |
| Mean Well NDR-480-24 G (Ground / PE)                     |                  | G      | 0  | 0  | 0  | 0  | 1   | Panel Ground Bus                                           | Equipment grounding conductor; bonds PSU chassis to panel ground bus and ultimately to building earth                                      | PSU G stud -> panel ground bus. Green or Green/Yellow wire per NFPA 79.                                                                                                               |
| Mean Well NDR-480-24 +V to EURO S4LH (Pushbutton Common) |                  | +V     | 0  | 0  | 0  | 0  | 1   | EURO S4LH Terminal Block -> Pushbutton Commons             | Fused +24V distribution to operator pushbutton common via S4LH                                                                             | +V -> 1 A fast-blow fuse (EURO S4LH terminal) -> distributes to common side of all illuminated pushbuttons (Forward S2PR-P3G, Reverse S2PR-P3Y, Run 1/2 S2PR-P3B x2, Reset S2PR-P3R). |
| Mean Well NDR-480-24 +V to E22B1 E-Stop Circuit          |                  | +V     | 0  | 0  | 0  | 0  | 1   | Cutler-Hammer E22B1 NC contact + SD-N35 contactor coil A1  | Direct +24V (no fuse) into E-stop safety chain                                                                                             | +V -> E22B1 NC contact -> SD-N35 A1 (contactor coil). PLC X4 taps this line between E22B1 and A1 to monitor E-stop state.                                                             |



| Point                | Description                    | Origin | DO | DI | AO | AI | Pwr | Destination Address                                     | Destination Description                                                                              | Notes                                                                                                                                                                     |
|----------------------|--------------------------------|--------|----|----|----|----|-----|---------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mean Well NDR-480-24 | +V to VFD FA (Jam Signal)      | +V     | 0  | 0  | 0  | 0  | 1   | VFD Huanyang HY02D211B-T FA terminal                    | Direct +24V (no fuse) into VFD jam signal circuit                                                    | +V -> VFD FA. When VFD fault relay closes (PD052=02), FA-FC bridges and delivers +24V to PLC X5.                                                                          |
| Mean Well NDR-480-24 | +V to F2-08AD-1 (Module Power) | +V     | 0  | 0  | 0  | 0  | 1   | F2-08AD-1 Analog Input Module +24V terminal             | Direct module power                                                                                  | +V -> F2-08AD-1 +24V terminal. No fuse on this branch.                                                                                                                    |
| Mean Well NDR-480-24 | +V to F2-08DA-2 (Module Power) | +V     | 0  | 0  | 0  | 0  | 1   | F2-08DA-2 Analog Output Module +24V terminal            | Direct module power                                                                                  | +V -> F2-08DA-2 +24V terminal. No fuse on this branch.                                                                                                                    |
| Mean Well NDR-480-24 | +V to F2-04RTD (Module Power)  | +V     | 0  | 0  | 0  | 0  | 1   | F2-04RTD RTD Input Module +24V terminal                 | Direct module power                                                                                  | +V -> F2-04RTD +24V terminal. No fuse on this branch.                                                                                                                     |
| Mean Well NDR-480-24 | +V to Indicator LEDs           | +V     | 0  | 0  | 0  | 0  | 1   | LED indicator (+) terminals (SA-LDG/Y/B/B/R, Wamco 525) | Direct +24V to (+) side of all illuminated push-button LED blocks and the separate Wamco error light | +V -> SA-LDG (+), SA-LDY (+), SA-LDB (+) x2, SA-LDR (+) (when wired), Wamco 525 (+). LED (-) sides return to D2-08TR relay outputs Y0-Y4 which sink to -V when energized. |
| Mean Well NDR-480-24 | -V to SD-N35 Contactor Coil A2 | -V     | 0  | 0  | 0  | 0  | 1   | Mitsubishi SD-N35 Contactor A2 (coil return)            | Contactor coil return through E-stop chain                                                           | -V -> SD-N35 A2 terminal. Coil energizes when E22B1 NC closes (E-stop not pressed) and +V reaches A1.                                                                     |
| Mean Well NDR-480-24 | -V to D2-08ND3 (Input Common)  | -V     | 0  | 0  | 0  | 0  | 1   | D2-08ND3 Discrete Input Module Common (C)               | Module input common for sourcing mode operation                                                      | -V -> D2-08ND3 C terminal. Sourcing mode reference - X inputs read HIGH when +24V is applied.                                                                             |

| Point                                                                      | Description | Origin | DO | DI | AO | AI | Pwr                                        | Destination Address                                                | Destination Description                                                                                                                                    | Notes |
|----------------------------------------------------------------------------|-------------|--------|----|----|----|----|--------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Mean Well NDR-480-24 -V to D2-08TR (Relay Common)                          | -V          | 0      | 0  | 0  | 0  | 1  | D2-08TR Relay Output Module Common (C)     | Relay common for sinking mode operation                            | -V -> D2-08TR C terminal. When any Y0-Y7 relay closes, that output terminal connects to -V (sinks current from the load).                                  |       |
| Mean Well NDR-480-24 -V to F2-08AD-1 (Module Power Return)                 | -V          | 0      | 0  | 0  | 0  | 1  | F2-08AD-1 Analog Input Module 0V terminal  | Module power return and analog signal common                       | -V -> F2-08AD-1 0V terminal.                                                                                                                               |       |
| Mean Well NDR-480-24 -V to F2-08DA-2 (Module Power Return / Analog Common) | -V          | 0      | 0  | 0  | 0  | 1  | F2-08DA-2 Analog Output Module 0V terminal | Module power return and analog signal reference for all 8 channels | -V -> F2-08DA-2 0V terminal. Single 0V terminal also serves as the analog signal common.                                                                   |       |
| Mean Well NDR-480-24 -V to F2-04RTD (Module Power Return)                  | -V          | 0      | 0  | 0  | 0  | 1  | F2-04RTD RTD Input Module 0V terminal      | Module power return                                                | -V -> F2-04RTD 0V terminal.                                                                                                                                |       |
| Mean Well NDR-480-24 -V to VFD DCM (Digital Common)                        | -V          | 0      | 0  | 0  | 0  | 1  | VFD Huanyang HY02D211B-T DCM terminal      | VFD digital input common reference                                 | -V -> VFD DCM. Provides 0V reference for FOR/REV/RST inputs (sinking/NPN mode). Closes the loop when D2-08TR relays Y5/Y6/Y7 pull those VFD inputs to DCM. |       |
| Mean Well NDR-480-24 -V to VFD ACM (Analog Common)                         | -V          | 0      | 0  | 0  | 0  | 1  | VFD Huanyang HY02D211B-T ACM terminal      | VFD analog signal common reference                                 | -V -> VFD ACM. Shared 0V reference for the analog speed command (VI from F2-08DA-2) and the analog current monitor (VO to F2-08AD-1).                      |       |
| Mean Well NDR-480-24 +V to HMI (EA1-T4CL)                                  | +V          | 0      | 0  | 0  | 0  | 1  | C-More EA1-T4CL HMI +VDC input             | HMI panel power                                                    | +V -> HMI +VDC. 24 VDC supply for the C-More Micro-Graphic panel (replaces former Rhino PSB12-030-P).                                                      |       |

| Point                                     | Descrip-<br>tion | Origin | DO | DI | AO | AI | Pwr | Destination<br>Address          | Destination<br>Descrip-<br>tion | Notes                                                         |
|-------------------------------------------|------------------|--------|----|----|----|----|-----|---------------------------------|---------------------------------|---------------------------------------------------------------|
| Mean Well NDR-480-24 -V to HMI (EA1-T4CL) |                  | -V     | 0  | 0  | 0  | 0  | 1   | C-More EA1-T4CL HMI 0VDC return | HMI panel power return          | -V -> HMI 0VDC. System DC common reference for the HMI panel. |
| Total Points                              |                  |        | 0  |    |    |    |     |                                 |                                 |                                                               |

## Sheet: Direct Logic 205 D2-09B-1

| Point                                                   | Description | Origin | DO | DI | AO | AI | Pwr | Destination Address | Destination Description                         | Notes                                                                                                                                                                    |
|---------------------------------------------------------|-------------|--------|----|----|----|----|-----|---------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PLC Rack (Direct-LOGIC 205) – Base                      |             | Base   | 0  | 0  | 0  | 0  | 1   | D2-09B-1            | Base / Power Unit                               | Rack base unit providing backplane power and I/O bus to all installed modules. Supports up to 9 I/O slots.                                                               |
| PLC Rack (Direct-LOGIC 205) – Line Input                |             | L      | 0  | 0  | 0  | 0  | 1   | L                   | MORSETTITALIA EURO S10H-5H Fuse Block           | Line from wall -> passes through power switch (closed = line passes) -> enters EURO S10H-5H fuse block with 1A fast blow fuse (PLC branch) -> enters PLC L terminal      |
| PLC Rack (Direct-LOGIC 205) – Neutral Input             |             | N      | 0  | 0  | 0  | 0  | 1   | N                   | Neutral Bus                                     | Neutral from wall -> directly connected to PLC N terminal                                                                                                                |
| PLC Rack (Direct-LOGIC 205) – Ground (Protective Earth) |             | G      | 0  | 0  | 0  | 0  | 1   | G                   | Ground Bus                                      | Protective earth -> connected to PLC ground terminal / chassis bond                                                                                                      |
| PLC Rack (Direct-LOGIC 205) – Slot 1                    |             | Slot 1 | 0  | 0  | 0  | 0  | 0   | H2-DM1E             | CPU / Ethernet Module                           | CPU module with built-in Ethernet port. Executes the ladder logic program and handles network communications.                                                            |
| PLC Rack (Direct-LOGIC 205) – Slot 2                    |             | Slot 2 | 0  | 8  | 0  | 0  | 0   | D2-08ND3            | 8-Point Discrete Input Module                   | 8-channel 24VDC discrete (digital) input module. Accepts sink/source wiring. Used for reading on/off signals such as pushbuttons, limit switches, and proximity sensors. |
| PLC Rack (Direct-LOGIC 205) – Slot 3                    |             | Slot 3 | 0  | 0  | 0  | 8  | 0   | F2-08AD-1           | 8-Channel Analog Current Input Module (4-20 mA) | 8-channel current input module. Used for motor current monitoring (CH1+ to VFD VO terminal, scaled 0-10V from VFD).                                                      |

| Point                                | Description | Origin | DO | DI | AO | AI | Pwr       | Destination Address                             | Destination Description                                                                                                                                      | Notes |
|--------------------------------------|-------------|--------|----|----|----|----|-----------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| PLC Rack (Direct-LOGIC 205) – Slot 4 | Slot 4      | 0      | 0  | 0  | 4  | 0  | F2-04RTD  | 4-Channel RTD Input Module                      | 4-channel RTD (Resistance Temperature Detector) input module. Compatible with PT100/PT1000 and other RTD types. Used for precision temperature measurement.  |       |
| PLC Rack (Direct-LOGIC 205) – Slot 5 | Slot 5      | 8      | 0  | 0  | 0  | 0  | D2-08TR   | 8-Point Relay Output Module                     | 8-channel relay (dry contact) output module. Common to PSU -V (sinking mode). Drives indicator LEDs (Y0-Y4) and VFD digital inputs (Y5=RST, Y6=FOR, Y7=REV). |       |
| PLC Rack (Direct-LOGIC 205) – Slot 6 | Slot 6      | 0      | 0  | 8  | 0  | 0  | F2-08DA-2 | 8-Channel Analog Voltage Output Module (0-10 V) | 8-channel voltage output module. CH1 (+V1) provides VFD speed reference to VI terminal (0-10V scaled to frequency).                                          |       |
| PLC Rack (Direct-LOGIC 205) – Slot 7 | Slot 7      | —      | —  | —  | —  | —  | D2-Fill   | Fill Panel                                      | Blank fill panel. No I/O. Slot reserved for future expansion.                                                                                                |       |
| PLC Rack (Direct-LOGIC 205) – Slot 8 | Slot 8      | —      | —  | —  | —  | —  | D2-Fill   | Fill Panel                                      | Blank fill panel. No I/O. Slot reserved for future expansion.                                                                                                |       |
| PLC Rack (Direct-LOGIC 205) – Slot 9 | Slot 9      | —      | —  | —  | —  | —  | D2-Fill   | Fill Panel                                      | Blank fill panel. No I/O. Slot reserved for future expansion.                                                                                                |       |

## Sheet: DirectLOGIC CPU (H2-DM1E)

| Point | Description                       | Origin                          | DO | DI | AO | AI | Pwr | Destination Address            | Destination Description                            | Notes                                                                                                                                                                                   |
|-------|-----------------------------------|---------------------------------|----|----|----|----|-----|--------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | Power and communication interface | Backplane Connector (Left Edge) | 0  | 0  | 0  | 0  | 1   | D2-09B-1 Base Slot 1 Connector | Provides power and backplane communication         | –                                                                                                                                                                                       |
|       | Ground reference                  | Backplane GND                   | 0  | 0  | 0  | 0  | 1   | D2-09B-1 Base Ground Bus       | Common ground for logic power                      | –                                                                                                                                                                                       |
|       | HMI communication port            | CPU Comm Port                   | 0  | 0  | 0  | 0  | 0   | HMI EA1-T4CL COM Port          | 4” C-More Micro-Graphic panel (operator interface) | CPU communication port links to the C-More HMI (see HMI sheet). Match baud, parity, and stop bits on both ends. Carries operator commands to the PLC and status/data back to the panel. |
|       |                                   | Total Points                    | 0  | 0  | 0  | 0  | 2   |                                |                                                    |                                                                                                                                                                                         |

# Sheet: Output Module (D2-08TA)

| Point                    | Descrip-<br>tion | Origin          | DO | DI | AO | AI | Pwr | Destination<br>Address    | Destination   Descrip-<br>tion                                              | Notes                                |
|--------------------------|------------------|-----------------|----|----|----|----|-----|---------------------------|-----------------------------------------------------------------------------|--------------------------------------|
| Output   Common<br>(Hot) |                  | C               | 1  | 0  | 0  | 0  | 1   | 120 V Hot Bus             | ZipLink   Terminal   Block<br>(ZL-RTB20-OUT). 1A in-<br>line slow blow fuse | 120 V hot feed shared by all outputs |
| Output 0                 |                  | OUT0            | 1  | 0  | 0  | 0  | 0   | A1                        | TX   Contactor   (GH15-<br>BN)                                              | Drives TX coil via ZipLink           |
| Output 1                 |                  | OUT1            | 1  | 0  | 0  | 0  | 0   | A1                        | BUS   Contactor   (GH15-<br>BN)                                             | Drives BUS coil via ZipLink          |
| Output 2                 |                  | OUT2            | 1  | 0  | 0  | 0  | 0   | A1                        | F1 Contactor (GH15-BN)                                                      | Drives F1 coil via ZipLink           |
| Output 3                 |                  | OUT3            | 1  | 0  | 0  | 0  | 0   | 95                        | Thermal   Overload   Relay<br>(RTD32-180)                                   | Feeds NC contact 95 → 96 → F2 coil   |
| Module Power Feed        |                  | Backplane       | 0  | 0  | 0  | 0  | 1   | Slot   3   Connec-<br>tor | PLC base backplane (120<br>VAC)                                             | No external wiring                   |
|                          |                  | Total<br>Points | 5  | 0  | 0  | 0  | 2   |                           |                                                                             |                                      |

# Sheet: HMI (EA1-T4CL)

| Point                | Descrip-<br>tion | Origin | DO | DI | AO | AI | Pwr    | Destination<br>Address | Destination<br>Descrip-<br>tion | Notes                                                                                                        |
|----------------------|------------------|--------|----|----|----|----|--------|------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------|
| HMI Serial Comm Port | COM Port         | 0      | 0  | 0  | 0  | 0  | 0      | Serial Port            | H2-DM1E                         | Serial link between PLC CPU and C-More HMI. Configure PLC and HMI ports for matching baud, parity, stop bits |
| HMI Power +V         | +VDC             | 0      | 0  | 0  | 0  | 1  | +V     | Mean Well NDR-480-24   |                                 | HMI 24 VDC input powered from Mean Well NDR-480-24 +V rail                                                   |
| HMI Power 0 V        | 0VDC             | 0      | 0  | 0  | 0  | 1  | 0V     | Mean Well NDR-480-24   |                                 | 0 VDC return to Mean Well NDR-480-24 -V rail (system DC common)                                              |
| HMI Ground           | Ground           | 0      | 0  | 0  | 0  | 1  | Ground | Ground                 |                                 |                                                                                                              |
|                      | Total Points     | 0      | 0  | 0  | 0  | 3  |        |                        |                                 |                                                                                                              |



## Sheet: Output Module (F2-02DAS-2)

| Point             | Descrip-<br>tion | Origin          | DO | DI | AO | AI | Pwr | Destination<br>Address  | Destination<br>Descrip-<br>tion                     | Notes                                                                |
|-------------------|------------------|-----------------|----|----|----|----|-----|-------------------------|-----------------------------------------------------|----------------------------------------------------------------------|
| Analog<br>(CH1)   | Common           | CH1-V           | 0  | 0  | 1  | 0  | 0   | CM                      | VFD (GS1-20P2) through<br>Ziplink ZL-RTB20          | 0V1 and CH1-V are internally shorted                                 |
| Analog<br>CH1     | Output           | CH1+V           | 0  | 0  | 1  | 0  | 0   | AL                      | VFD (GS1-20P2) though<br>Ziplink ZL-RTB20           | 0–10 V speed command to VFD                                          |
| Module Logic<br>V | +24              | +V1             | 0  | 0  | 0  | 0  | 1   | VDC +24V                | PLC Base (D2-09B-1)<br>through ZipLink ZL-<br>RTB20 | Powers analog output circuitry, also fused<br>with 1A fast blow fuse |
| Module Logic      | 0 V              | 0V1             | 0  | 0  | 0  | 0  | 1   | VDC – 0V                | PLC Base (D2-09B-1)<br>through ZipLink ZL-<br>RTB20 | Common return for module logic                                       |
| Slot Power Feed   |                  | Backplane       | 0  | 0  | 0  | 0  | 1   | PLC Base Back-<br>plane | Internal slot power                                 | No external wiring                                                   |
|                   |                  | Total<br>Points | 0  | 0  | 2  | 0  | 3   |                         |                                                     |                                                                      |

## Sheet: Input Module (D2-08ND3)

| Point       | Descrip-<br>tion | Origin | DO | DI | AO | AI | Pwr | Destination<br>Address        | Destination<br>Descrip-<br>tion                                                                                   | Notes                                                                                  |
|-------------|------------------|--------|----|----|----|----|-----|-------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| D2-08ND3 X0 |                  | X0     | 0  | 1  | 0  | 0  | 0   | Forward Push-button           | Forward button S2PR-P3G with Contact block SA-CA with LED module SA-LDG                                           | Shared button +24V -> NO contact -> X0.                                                |
| D2-08ND3 X1 |                  | X1     | 0  | 1  | 0  | 0  | 0   | Reverse Push-button           | Reverse button S2PR-P3Y with Contact block SA-CA with LED module SA-LDY                                           | Shared button +24V -> NO contact -> X1.                                                |
| D2-08ND3 X2 |                  | X2     | 0  | 1  | 0  | 0  | 0   | Run 1 / Dead-man 1 Pushbutton | Run 1 button S2PR-P3B with Contact block SA-CA with LED module SA-LDB                                             | Shared button +24V -> NO contact -> X2.                                                |
| D2-08ND3 X3 |                  | X3     | 0  | 1  | 0  | 0  | 0   | Run 2 / Dead-man 2 Pushbutton | Run 2 button S2PR-P3B with Contact block SA-CA with LED module SA-LDB                                             | Shared button +24V -> NO contact -> X3.                                                |
| D2-08ND3 X4 |                  | X4     | 0  | 1  | 0  | 0  | 0   | E-Stop Button Monitor         | E-stop button E22B1 with NC contact block, wired in series with Mitsubishi SD-N35 contactor coil                  | PSU +V -> E22B1 NC -> SD-N35 A1 (X4 tapped between E22B1 and A1). SD-N35 A2 -> PSU -V. |
| D2-08ND3 X5 |                  | X5     | 0  | 1  | 0  | 0  | 0   | VFD Fault Relay               | VFD fault output Huanyang HY02D211B-T internal SPDT relay (FA/FC/FB), programmed as fault indication via PD052=02 | PSU +V -> VFD FA. VFD FC -> X5. FB unused.                                             |

| Point                     | Description | Origin | DO | DI | AO | AI | Pwr | Destination Address                   | Destination Description                                                                                      | Notes                                                                                                                                                                                                                                                    |
|---------------------------|-------------|--------|----|----|----|----|-----|---------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D2-08ND3 X6               |             | X6     | 0  | 1  | 0  | 0  | 0   | Reserved - Reset Pushbutton (planned) | Reset button S2PR-P3R with Contact block SA-CA with LED module SA-LDR                                        | Unwired. Planned: shared button +24V -> NO contact -> X6 (same topology as X0-X3).                                                                                                                                                                       |
| D2-08ND3 X7               |             | X7     | 0  | 1  | 0  | 0  | 0   | Spare                                 | Unused                                                                                                       | Unwired.                                                                                                                                                                                                                                                 |
| D2-08ND3 Input Common (C) |             | C      | 0  | 0  | 0  | 0  | 1   | Mean Well NDR-480-24 -V               | DC power supply 0V return rail. Both top-row and bottom-row C terminals on the D2-08ND3 are internally tied. | C -> PSU -V. Module wired in sourcing mode. Shared +24V button supply is fed from PSU +V through a 1 A fast-blow fuse (EURO S10H-5H holder) into the EURO S4LH distribution terminal, which provides the common +24V to X0/X1/X2/X3 (and X6 when wired). |
| Total Points              |             |        | 0  | 8  | 0  | 0  | 1   |                                       |                                                                                                              |                                                                                                                                                                                                                                                          |

## Sheet: Analog Input (F2-08AD-1)

| Point        | Description                 | Origin | DO | DI | AO | AI | Pwr | Destination Address           | Destination Description                                                                                                                   | Notes                                                                                             |
|--------------|-----------------------------|--------|----|----|----|----|-----|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| F2-08AD-1    | +24V<br>(Module Power)      | +24V   | 0  | 0  | 0  | 0  | 1   | Mean Well<br>NDR-480-24<br>+V | DC power supply +24V rail                                                                                                                 | PSU +V -> Module +24V terminal directly. No fuse on this branch.                                  |
| F2-08AD-1    | 0V<br>(Module Power Return) | 0V     | 0  | 0  | 0  | 0  | 1   | Mean Well<br>NDR-480-24 -V    | DC power supply 0V return rail. Shared with VFD DCM and ACM (all tied to PSU -V).                                                         | PSU -V -> Module 0V terminal. Also serves as the common reference for channel inputs.             |
| F2-08AD-1    | CH1+<br>(Channel 1 Signal)  | CH1+   | 0  | 0  | 0  | 1  | 0   | VFD Motor<br>Current Monitor  | VFD analog output<br>Huanyang HY02D211B-T<br>VO terminal, configured for output current via PD054=1 (0-10V proportional to motor current) | VFD VO -> Module CH1+ terminal. Signal return reference via PSU -V (VFD ACM also tied to PSU -V). |
| F2-08AD-1    | CH2                         | CH2    | 0  | 0  | 0  | 1  | 0   | Spare                         | Unused channel                                                                                                                            | Unwired.                                                                                          |
| F2-08AD-1    | CH3                         | CH3    | 0  | 0  | 0  | 1  | 0   | Spare                         | Unused channel                                                                                                                            | Unwired.                                                                                          |
| F2-08AD-1    | CH4                         | CH4    | 0  | 0  | 0  | 1  | 0   | Spare                         | Unused channel                                                                                                                            | Unwired.                                                                                          |
| F2-08AD-1    | CH5                         | CH5    | 0  | 0  | 0  | 1  | 0   | Spare                         | Unused channel                                                                                                                            | Unwired.                                                                                          |
| F2-08AD-1    | CH6                         | CH6    | 0  | 0  | 0  | 1  | 0   | Spare                         | Unused channel                                                                                                                            | Unwired.                                                                                          |
| F2-08AD-1    | CH7                         | CH7    | 0  | 0  | 0  | 1  | 0   | Spare                         | Unused channel                                                                                                                            | Unwired.                                                                                          |
| F2-08AD-1    | CH8                         | CH8    | 0  | 0  | 0  | 1  | 0   | Spare                         | Unused channel                                                                                                                            | Unwired.                                                                                          |
| Total Points |                             |        | 0  | 0  | 0  | 8  | 2   |                               |                                                                                                                                           |                                                                                                   |

## Sheet: RTD Input (F2-04RTD)

| Point        | Descrip-<br>tion               | Origin | DO | DI | AO | AI | Pwr | Destination<br>Address        | Destination<br>Descrip-<br>tion     | Notes                                                                                                          |
|--------------|--------------------------------|--------|----|----|----|----|-----|-------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------|
| F2-04RTD     | +24V<br>(Module Power)         | +24V   | 0  | 0  | 0  | 0  | 1   | Mean Well<br>NDR-480-24<br>+V | DC power supply +24V<br>rail        | PSU +V -> Module +24V terminal. Mod-<br>ule installed in rack but no channels currently<br>used in this build. |
| F2-04RTD     | 0V<br>(Module Power<br>Return) | 0V     | 0  | 0  | 0  | 0  | 1   | Mean Well<br>NDR-480-24 -V    | DC power supply 0V re-<br>turn rail | PSU -V -> Module 0V terminal.                                                                                  |
| F2-04RTD     | CH1                            | CH1    | 0  | 0  | 0  | 1  | 0   | Spare                         | Unused RTD input chan-<br>nel       | Unwired. Available for future RTD tempera-<br>ture sensor.                                                     |
| F2-04RTD     | CH2                            | CH2    | 0  | 0  | 0  | 1  | 0   | Spare                         | Unused RTD input chan-<br>nel       | Unwired.                                                                                                       |
| F2-04RTD     | CH3                            | CH3    | 0  | 0  | 0  | 1  | 0   | Spare                         | Unused RTD input chan-<br>nel       | Unwired.                                                                                                       |
| F2-04RTD     | CH4                            | CH4    | 0  | 0  | 0  | 1  | 0   | Spare                         | Unused RTD input chan-<br>nel       | Unwired.                                                                                                       |
| Total Points |                                |        | 0  | 0  | 0  | 4  | 2   |                               |                                     |                                                                                                                |

## Sheet: Relay Output (D2-08TR)

| Point   | Descrip-<br>tion | Origin | DO | DI | AO | AI | Pwr | Destination<br>Address                  | Destination<br>Descrip-<br>tion                                                                                        | Notes                                                                                                                                      |
|---------|------------------|--------|----|----|----|----|-----|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| D2-08TR | Y0               | Y0     | 1  | 0  | 0  | 0  | 0   | Forward Indica-<br>tor LED              | LED module SA-LDG<br>(green, 12-24 VDC)<br>integrated in Forward<br>pushbutton S2PR-P3G<br>(assembly also drives X0)   | PSU +V -> SA-LDG (+). SA-LDG (-) -> Y0<br>contact. Relay sinks Y0 to -V when energized,<br>lighting the LED.                               |
| D2-08TR | Y1               | Y1     | 1  | 0  | 0  | 0  | 0   | Reverse Indica-<br>tor LED              | LED module SA-LDY<br>(yellow, 12-24 VDC)<br>integrated in Reverse<br>pushbutton S2PR-P3Y<br>(assembly also drives X1)  | PSU +V -> SA-LDY (+). SA-LDY (-) -> Y1<br>contact. Relay sinks Y1 to -V when energized.                                                    |
| D2-08TR | Y2               | Y2     | 1  | 0  | 0  | 0  | 0   | Run 1 / Dead-<br>man 1 Indicator<br>LED | LED module SA-LDB<br>(blue, 12-24 VDC)<br>integrated in Run 1 push-<br>button S2PR-P3B #1<br>(assembly also drives X2) | PSU +V -> SA-LDB (+). SA-LDB (-) -> Y2<br>contact.                                                                                         |
| D2-08TR | Y3               | Y3     | 1  | 0  | 0  | 0  | 0   | Run 2 / Dead-<br>man 2 Indicator<br>LED | LED module SA-LDB<br>(blue, 12-24 VDC)<br>integrated in Run 2 push-<br>button S2PR-P3B #2<br>(assembly also drives X3) | PSU +V -> SA-LDB (+). SA-LDB (-) -> Y3<br>contact.                                                                                         |
| D2-08TR | Y4               | Y4     | 1  | 0  | 0  | 0  | 0   | Error / Fault In-<br>dicator            | Wamco model 525 panel-<br>mount LED indicator (28<br>VDC, 0.6 W)                                                       | PSU +V -> Wamco 525 (+). Wamco 525 (-)<br>-> Y4 contact. 28V-rated indicator slightly<br>under-driven at 24V (acceptable, ~21 mA<br>draw). |

| Point        | Descrip-<br>tion    | Origin | DO | DI | AO | AI | Pwr | Destination<br>Address     | Destination<br>Descrip-<br>tion                                                     | Notes                                                                                                                                              |
|--------------|---------------------|--------|----|----|----|----|-----|----------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| D2-08TR      | Y5                  | Y5     | 1  | 0  | 0  | 0  | 0   | VFD Fault Re-<br>set       | Huanyang HY02D211B-<br>T VFD RST terminal<br>(multi-input, sink-<br>ing/NPN mode)   | Y5 -> VFD RST. When Y5 energized, relay<br>closes Y5 to common (PSU -V), pulling RST<br>input to DCM. ~500 ms pulse during PLC un-<br>jam routine. |
| D2-08TR      | Y6                  | Y6     | 1  | 0  | 0  | 0  | 0   | VFD Forward<br>Run         | Huanyang HY02D211B-<br>T VFD FOR termi-<br>nal (multi-input, sink-<br>ing/NPN mode) | Y6 -> VFD FOR. When Y6 energized, pulls<br>FOR to DCM. Forward shredding.                                                                          |
| D2-08TR      | Y7                  | Y7     | 1  | 0  | 0  | 0  | 0   | VFD Reverse<br>Run         | Huanyang HY02D211B-<br>T VFD REV termi-<br>nal (multi-input, sink-<br>ing/NPN mode) | Y7 -> VFD REV. When Y7 energized, pulls<br>REV to DCM. Used during PLC unjam rou-<br>tine.                                                         |
| D2-08TR      | Relay<br>Common (C) | C      | 0  | 0  | 0  | 0  | 1   | Mean Well<br>NDR-480-24 -V | DC power supply 0V re-<br>turn rail                                                 | C-> PSU -V. Sinking mode: when any Y0-Y7<br>closes, that terminal connects to -V.                                                                  |
| Total Points |                     |        | 8  | 0  | 0  | 0  | 1   |                            |                                                                                     |                                                                                                                                                    |

## Sheet: Analog Output (F2-08DA-2)

| Point        | Description                              | Origin | DO | DI | AO | AI | Pwr | Destination Address        | Destination Description                                                                                                        | Notes                                                                                                                           |
|--------------|------------------------------------------|--------|----|----|----|----|-----|----------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| F2-08DA-2    | +24V (Module Power)                      | +24V   | 0  | 0  | 0  | 0  | 1   | Mean Well NDR-480-24 +V    | DC power supply +24V rail                                                                                                      | PSU +V -> Module +24V terminal directly. No fuse on this branch.                                                                |
| F2-08DA-2    | 0V (Module Power Return / Analog Common) | 0V     | 0  | 0  | 0  | 0  | 1   | Mean Well NDR-480-24 -V    | DC power supply 0V return rail. Single 0V terminal serves both as power return and analog signal reference for all 8 channels. | PSU -V -> Module 0V terminal. VFD ACM is also tied to PSU -V, so the analog signal return is shared through the common 0V rail. |
| F2-08DA-2    | +V1 (Channel 1 Signal)                   | +V1    | 0  | 0  | 1  | 0  | 0   | VFD Analog Speed Reference | Huanyang HY02D211B-T VFD VI terminal (analog voltage input, 0-10 V), used as frequency/RPM reference. Requires PD002=1.        | +V1 -> VFD VI. 0-10 V signal scaled to commanded frequency: 0 V = 0 Hz, 10 V = max frequency.                                   |
| F2-08DA-2    | +V2                                      | +V2    | 0  | 0  | 1  | 0  | 0   | Spare                      | Unused channel                                                                                                                 | Unwired.                                                                                                                        |
| F2-08DA-2    | +V3                                      | +V3    | 0  | 0  | 1  | 0  | 0   | Spare                      | Unused channel                                                                                                                 | Unwired.                                                                                                                        |
| F2-08DA-2    | +V4                                      | +V4    | 0  | 0  | 1  | 0  | 0   | Spare                      | Unused channel                                                                                                                 | Unwired.                                                                                                                        |
| F2-08DA-2    | +V5                                      | +V5    | 0  | 0  | 1  | 0  | 0   | Spare                      | Unused channel                                                                                                                 | Unwired.                                                                                                                        |
| F2-08DA-2    | +V6                                      | +V6    | 0  | 0  | 1  | 0  | 0   | Spare                      | Unused channel                                                                                                                 | Unwired.                                                                                                                        |
| F2-08DA-2    | +V7                                      | +V7    | 0  | 0  | 1  | 0  | 0   | Spare                      | Unused channel                                                                                                                 | Unwired.                                                                                                                        |
| F2-08DA-2    | +V8                                      | +V8    | 0  | 0  | 1  | 0  | 0   | Spare                      | Unused channel                                                                                                                 | Unwired.                                                                                                                        |
| Total Points |                                          |        | 0  | 0  | 8  | 0  | 2   |                            |                                                                                                                                |                                                                                                                                 |



## Sheet: VFD (Huanyang HY02D211B-T)

| Point                                                        | Descrip-<br>tion | Origin | DO | DI | AO | AI | Pwr | Destination<br>Address                                                                                          | Destination<br>Descrip-<br>tion                                                                                                                                                                                          | Notes                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------|------------------|--------|----|----|----|----|-----|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VFD R (AC Line<br>Hot Input)                                 | R                |        | 0  | 0  | 0  | 0  | 1   | AC Mains Hot<br>via main discon-<br>nect and 30 A<br>breaker                                                    | 120 VAC line conductor<br>from wall outlet, routed<br>through panel main dis-<br>connect and a 30 A<br>C-curve DIN-rail circuit<br>breaker                                                                               | Wall AC L -> disconnect -> 30 A breaker -><br>VFD R. Black wire per NFPA 79.                                                                                                                                                                                                              |
| VFD T (AC Neu-<br>tral Input via Con-<br>tactor and Breaker) | T                |        | 0  | 0  | 0  | 0  | 1   | AC Mains Neu-<br>tral via Phoenix<br>Contact TMC 8<br>3C 15A breaker<br>and Mitsubishi<br>SD-N35 contac-<br>tor | Neutral leg switched<br>through the safety-<br>disconnect contactor.<br>As-built path: Wall N -><br>Phoenix Contact TMC<br>8 3C 15A breaker (P/N<br>2907618) -> SD-N35 T1<br>-> contactor pole -><br>SD-N35 L1 -> VFD T. | AS-BUILT: neutral leg is switched by the con-<br>tactor instead of the hot leg. Note: NFPA 79<br>/ NEC standard practice is to switch the HOT<br>leg. Review for final report. When E-stop is<br>pressed, SD-N35 coil drops -> contact opens<br>-> VFD T loses neutral -> VFD shuts down. |
| VFD Ground (PE)                                              | Ground           | Ground | 0  | 0  | 0  | 0  | 1   | Panel Ground<br>Bus                                                                                             | Equipment grounding<br>conductor; bonds VFD<br>chassis to panel ground<br>bus                                                                                                                                            | VFD ground stud -> panel ground bus. Green<br>or Green/Yellow wire.                                                                                                                                                                                                                       |
| VFD U (Motor<br>Phase 1 Output)                              | U                |        | 0  | 0  | 0  | 0  | 1   | Banana Socket<br>(Blue) -> Motor<br>T1                                                                          | Panel-mount banana<br>socket (blue, 4 mm)<br>provides quick-disconnect<br>to GE 5KE182BC205B<br>motor T1                                                                                                                 | VFD U -> blue banana socket -> motor ter-<br>minal.                                                                                                                                                                                                                                       |

| Point   | Description             | Origin | DO | DI | AO | AI | Pwr | Destination Address                                 | Destination Description                                                                       | Notes                                           |
|---------|-------------------------|--------|----|----|----|----|-----|-----------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------|
| VFD V   | (Motor Phase 2 Output)  | V      | 0  | 0  | 0  | 0  | 1   | Banana Socket (White) -> Motor T2                   | Panel-mount banana socket (white, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T2 | VFD V -> white banana socket -> motor terminal. |
| VFD W   | (Motor Phase 3 Output)  | W      | 0  | 0  | 0  | 0  | 1   | Banana Socket (Red) -> Motor T3                     | Panel-mount banana socket (red, 4 mm) provides quick-disconnect to GE 5KE182BC205B motor T3   | VFD W -> red banana socket -> motor terminal.   |
| VFD P+  | (Brake Resistor +)      | P+     | 0  | 0  | 0  | 0  | 1   | Brake Resistor ATO APCS-300R30-AD (+ terminal)      | Dynamic braking resistor 300 W, 30 Ohms, aluminum-clad (ATO model APCS-300R30-AD)             | VFD P+ -> brake resistor terminal 1.            |
| VFD PR  | (Brake Resistor Return) | PR     | 0  | 0  | 0  | 0  | 1   | Brake Resistor ATO APCS-300R30-AD (return terminal) | Dynamic braking resistor (same component as P+ row)                                           | VFD PR -> brake resistor terminal 2.            |
| VFD UPF |                         | UPF    | 0  | 1  | 0  | 0  | 0   | Unused                                              | Multi-Output 2 (PD051), open-collector optocoupler                                            | Unwired.                                        |
| VFD DRV |                         | DRV    | 0  | 1  | 0  | 0  | 0   | Unused                                              | Multi-Output 1 (PD050), open-collector optocoupler                                            | Unwired.                                        |

| Point<br>Description              | Origin | DO | DI | AO | AI | Pwr | Destination<br>Address  | Destination<br>Description                                                  | Notes                                                                                            |
|-----------------------------------|--------|----|----|----|----|-----|-------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| VFD DCM (Digital Common, top row) | DCM    | 0  | 0  | 0  | 0  | 1   | Mean Well NDR-480-24 -V | DC power supply 0V return - shared reference for FOR/REV/RST sinking inputs | VFD DCM (top row) -> PSU -V. PSU -V also feeds D2-08TR relay common.                             |
| VFD SPL                           | SPL    | 0  | 1  | 0  | 0  | 0   | Unused                  | Multi-Input 6 - default low-speed select                                    | Unwired.                                                                                         |
| VFD SPM                           | SPM    | 0  | 1  | 0  | 0  | 0   | Unused                  | Multi-Input 5 - default middle-speed select                                 | Unwired.                                                                                         |
| VFD SPH                           | SPH    | 0  | 1  | 0  | 0  | 0   | Unused                  | Multi-Input 4 - default high-speed select                                   | Unwired.                                                                                         |
| VFD RST (Fault Reset Input)       | RST    | 0  | 1  | 0  | 0  | 0   | D2-08TR Y5              | PLC relay output (sinking NPN mode) driving fault reset                     | PLC Y5 -> VFD RST. Pulled to DCM (0V) when Y5 closes.                                            |
| VFD REV (Reverse Run Input)       | REV    | 0  | 1  | 0  | 0  | 0   | D2-08TR Y7              | PLC relay output (sinking NPN mode) driving reverse run command             | PLC Y7 -> VFD REV. Pulled to DCM (0V) when Y7 closes.                                            |
| VFD FOR (Forward Run Input)       | FOR    | 0  | 1  | 0  | 0  | 0   | D2-08TR Y6              | PLC relay output (sinking NPN mode) driving forward run command             | PLC Y6 -> VFD FOR. Pulled to DCM (0V) when Y6 closes.                                            |
| VFD ACM (Analog Common, top row)  | ACM    | 0  | 0  | 0  | 0  | 1   | Mean Well NDR-480-24 -V | DC power supply 0V return - shared reference for analog signals             | VFD ACM (top row) -> PSU -V. Shared 0V reference for VI (speed in) and VO (current monitor out). |
| VFD VO (Analog Output, 0-10 V)    | VO     | 0  | 0  | 1  | 0  | 0   | F2-08AD-1 CH1+          | PLC analog input module CH1, used as motor current monitor (PD054=1)        | VFD VO -> F2-08AD-1 CH1+. 0-10 V scaled to motor output current.                                 |
| VFD 10V                           | 10V    | 0  | 0  | 1  | 0  | 0   | Unused                  | +10 V reference output (for external potentiometer)                         | Unwired.                                                                                         |

| Point<br>Description                   | Origin       | DO | DI | AO | AI | Pwr | Destination<br>Address     | Destination<br>Description                                                             | Notes                                                                                          |
|----------------------------------------|--------------|----|----|----|----|-----|----------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| VFD FA (Multi-Output 3 Relay NO)       | FA           | 0  | 1  | 0  | 0  | 0   | Mean Well NDR-480-24 +V    | DC power supply +V rail                                                                | PSU +V -> VFD FA. No fuse on this branch. FA-FC closes when VFD trips on any fault (PD052=02). |
| VFD FC (Multi-Output 3 Relay Common)   | FC           | 0  | 1  | 0  | 0  | 0   | D2-08ND3 X5                | PLC discrete input X5 (jam signal)                                                     | VFD FC -> PLC X5. When FA-FC closes, +24V from FA reaches X5 (PLC reads HIGH = jam).           |
| VFD FB (Multi-Output 3 Relay NC)       | FB           | 0  | 1  | 0  | 0  | 0   | Unused                     | Normally-closed contact - not used in active-high NO configuration                     | Unwired.                                                                                       |
| VFD 24V (On-board 24V Supply)          | 24V          | 0  | 0  | 0  | 0  | 1   | Unused                     | Onboard 24 VDC auxiliary supply, max 200 mA                                            | Unwired. Panel uses external Mean Well NDR-480-24 instead.                                     |
| VFD DCM (bottom row)                   | DCM (bottom) | 0  | 0  | 0  | 0  | 1   | (same node as top-row DCM) | Internally tied to top-row DCM                                                         | Alternate landing point. Top-row DCM is used.                                                  |
| VFD 5V                                 | 5V           | 0  | 0  | 1  | 0  | 0   | Unused                     | +5 V reference output                                                                  | Unwired.                                                                                       |
| VFD ACM (bottom row)                   | ACM (bottom) | 0  | 0  | 0  | 0  | 1   | (same node as top-row ACM) | Internally tied to top-row ACM                                                         | Alternate landing point. Top-row ACM is used.                                                  |
| VFD AI (Analog Current Input, 4-20 mA) | AI           | 0  | 0  | 0  | 1  | 0   | Unused                     | Analog current input for speed reference. Speed command uses VI voltage input instead. | Unwired.                                                                                       |
| VFD VI (Analog Voltage Input, 0-10 V)  | VI           | 0  | 0  | 0  | 1  | 0   | F2-08DA-2 +V1              | PLC analog output module CH1, used as commanded frequency reference (PD002=1)          | F2-08DA-2 +V1 -> VFD VI. 0-10 V scaled to commanded frequency.                                 |
| VFD RS-                                | RS-          | 0  | 0  | 0  | 0  | 1   | Unused                     | RS-485 Modbus communication negative line                                              | Unwired. Modbus not used.                                                                      |

| Point        | Descrip-<br>tion | Origin | DO | DI | AO | AI | Pwr | Destination<br>Address | Destination<br>Descrip-<br>tion                | Notes    |
|--------------|------------------|--------|----|----|----|----|-----|------------------------|------------------------------------------------|----------|
| VFD RS+      |                  | RS+    | 0  | 0  | 0  | 0  | 1   | Unused                 | RS-485 Modbus communi-<br>cation positive line | Unwired. |
| Total Points |                  |        | 0  | 11 | 3  | 2  | 15  |                        |                                                |          |

## 16.3 Appendix C: PLC Ladder Logic — Reference

The complete 31-rung Do-more Designer ladder export is reproduced inline in Section 6.1 (PLC Ladder Logic Architecture), with each page of `electrical/Circuit_Logic.pdf` placed adjacent to the rung-by-rung walkthrough that describes its contents. The full PDF source is also retained at `electrical/Circuit_Logic.pdf` in the repository for archival and detailed inspection.

## 16.4 Appendix D: Wiring Diagrams

**[TODO: Insert exported PDFs from AutoCAD: power, VFD/motor, PLC/HMI, power distribution.]**

## 16.5 Appendix E: VFD Parameter Settings

The full programmed parameter set — code, function, value, and rationale — is given in Table 1 (Section 5.2, Motor and VFD). Motor values PD141–144 reflect the GE 5KE182BC205B test motor; the production motor is TBD and its nameplate will supersede these.

**Power / braking wiring:** R = AC hot through the SD-N35 contactor (and the 30 A branch breaker); T = AC neutral; U/V/W to motor via colored banana sockets; P+/PR to a 300 W, 30  $\Omega$  brake resistor (ATO APCS-300R30-AD) to absorb regenerative energy during reversal in the jam-clear cycle. Control wiring is summarized in Appendix B.

*[NOTE: Firmware-specific finding: PD052 = 12 (continue-running over-torque indication) did not fire the relay on our Huanyang revision; the working configuration is PD052 = 02 with PD123 = 3 (over-torque self-trips, which fires the fault relay and is read by the PLC on X5).]*

## 16.6 Appendix F: Full PLC Variable Reference

The compact key-terms table in Section 6.1 (PLC Ladder Logic Architecture) gives a one-phrase meaning for every tag used in the ladder. The table below is the full address-by-address reference: every X / Y / WX / WY / C / T / CT / V / ST point used in the program, with its electrical type or instruction kind and a longer description that names the source module, the ladder context, and any related rungs.

| Address | Name in Ladder | Type | Description                                          |
|---------|----------------|------|------------------------------------------------------|
| X0      | FORWARDS       | DI   | Forward pushbutton (green S2PR-P3G). HIGH = pressed. |

*continued on next page...*

| Address | Name in Ladder | Type | Description                                                                                                           |
|---------|----------------|------|-----------------------------------------------------------------------------------------------------------------------|
| X1      | REVERSE        | DI   | Reverse pushbutton (yellow S2PR-P3Y). HIGH = pressed.                                                                 |
| X2      | DEADMAN_A_OK   | DI   | Deadman button A (blue #1). Must be held for DEADMEN_OK.                                                              |
| X3      | DEADMAN_B_OK   | DI   | Deadman button B (blue #2). Must be held simultaneously with X2.                                                      |
| X4      | STOP_PB        | DI   | E-stop NC contact monitor. LOW when E-stop is pressed. Used as NOT X4 in rungs 17, 18.                                |
| X5      | OVERCURRENT    | DI   | VFD fault relay input (FA/FC, PD052=02). HIGH = VFD tripped. Triggers jam recovery sequence.                          |
| X7      | CAP_TOUCH_TRIP | DI   | Capacitive touch safety sensor (on-hand, wired). HIGH = safety zone breached. Immediately sets STOP_MOTOR, CTSI_TRIG. |
| Y0      | FWD_LED        | DO   | Forward indicator LED (green SA-LDG). ON when FORWARDS_SET (C8) is HIGH.                                              |
| Y1      | REVERSE_LED    | DO   | Reverse indicator LED (yellow SA-LDY). ON when REVERSE_SET (C9) is HIGH.                                              |
| Y2      | DMAN_A_LED     | DO   | Deadman A LED (blue #1). ON when (X2 OR C23) AND NOT Y4.                                                              |
| Y3      | DMAN_B_LED     | DO   | Deadman B LED (blue #2). ON when (X3 OR C24) AND NOT Y4.                                                              |
| Y4      | E_LIGHT        | DO   | Error / fault indicator (Wamco 525 panel LED). SET by E-stop open, CAP_TOUCH, and CT0.Done. RST by ERROR_RESET (C18). |

*continued on next page...*

| Address | Name in Ladder     | Type       | Description                                                                                                 |
|---------|--------------------|------------|-------------------------------------------------------------------------------------------------------------|
| Y5      | VFD_RESET          | DO         | VFD fault reset output (RST terminal). Energized while DEADMEN_OK AND X5 are both HIGH (rung 19).           |
| Y6      | VFD_RUN_FWD        | DO         | VFD forward run (FOR terminal). Energized by rung 9 when all permissives are met and no fault/pause active. |
| Y7      | VFD_RUN_RVS        | DO         | VFD reverse run (REV terminal). Energized by rung 10 via REVERSE_SET or unconditionally by UNJAM_R (C13).   |
| WX0     | Analog input word  | AI word    | Raw ADC from F2-08AD-1 CH1 (VFD VO, motor current). Input to rung 3 SCALE (0–4095 → 0–140).                 |
| WY0     | Analog output word | AO word    | Raw DAC to F2-08DA-2 CH1 (VFD VI, speed reference). Output of rung 2 SCALE (V2000 0–1600 → 0–4095).         |
| C0      | DEADMEN_OK         | BOOL       | HIGH when (X2 OR C23) AND (X3 OR C24). Both deadman paths held.                                             |
| C1      | HMI_STP_REQ        | BOOL       | HMI STOP button request. Rung 7: NOT C1 gates RUN_REQUEST_SET.                                              |
| C3      | PERMISSIVE_OK      | BOOL       | Mirrors DEADMEN_OK (C0). Used as interlock in rungs 7 and 8.                                                |
| C8      | FORWARDS_SET       | BOOL latch | Forward direction selected. SET by X0 (rung 13) or on first scan init (rung 4). RST by X1.                  |

*continued on next page...*



| Address | Name in Ladder   | Type       | Description                                                                                                                              |
|---------|------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------|
| C9      | REVERSE_SET      | BOOL latch | Reverse direction selected. SET by X1 (rung 14). RST by X0. RSTR clears C9–C30 on first scan.                                            |
| C10     | JAM_LATCH        | BOOL       | Output of T0 (OFFDTMR). HIGH while overcurrent is active; stays HIGH 2 s after X5 drops. Blocks Y6, drives T1.                           |
| C11     | init flag        | BOOL       | SET on first scan by \$FirstScan (rung 1 RST, rung 4 triggers init). Used to initialize FORWARDS_SET and clear C9–C30.                   |
| C12     | RUN_REQUEST      | BOOL latch | Motor run command. SET by rung 7, RST by rung 8 (NOT X4 or NOT PERMISSIVE_OK). Also shown as MOTOR_STATUS on HMI.                        |
| C13     | UNJAM_R          | BOOL       | Output of T1 (OFFDTMR, 2 s). HIGH while JAM_LATCH is HIGH; stays HIGH 2 s after JAM_LATCH drops. Energizes Y7 (reverse) unconditionally. |
| C15     | STOP_MOTOR       | BOOL latch | Hard stop flag. SET by CAP_TOUCH or CT0.Done (lock-out). RST by ERROR_RESET (C18). Blocks Y6 and Y7 in rungs 9, 10.                      |
| C16     | LOCKOUT_TMR_STRT | BOOL latch | Enables T2 accumulating timer. SET when UNJAM_R active and T2 not timing. RST when T2.DN fires (30 s accumulated).                       |
| C17     | PAUSE_MOTOR      | BOOL       | Output of T3 (OFFDTMR, 2 s). Triggered by T1.DN rising edge. Blocks Y6 for 2 s after reverse completes (coast-to-stop buffer).           |

*continued on next page...*

| Address | Name in Ladder   | Type       | Description                                                                                                                    |
|---------|------------------|------------|--------------------------------------------------------------------------------------------------------------------------------|
| C18     | ERROR_RESET      | BOOL       | Fault reset coil (from HMI ERROR_OFF button, Screen 3). RSTs STOP_MOTOR, E_LIGHT, JAMMED_MOTOR, CTSI_TRIG, CHANGE_SCREEN, CT0. |
| C20     | JAMMED_MOTOR     | BOOL latch | Lockout flag. SET when CT0.Done fires (3rd jam). Shown on Screen 3 SHREDDER JAM indicator. RST by ERROR_RESET.                 |
| C21     | CTSI_TRIG        | BOOL latch | Capacitive touch triggered. SET by X7 (rung 28) or ERROR_RESET path. Shown on Screen 3. RST by ERROR_RESET.                    |
| C22     | CHANGE_SCREEN    | BOOL latch | Triggers HMI screen change to error screen (Screen 3). Rung 29: sets V100=3 and starts T4 (3 s). T4.DN resets V100 to 0.       |
| C23     | TESTMAN_A_OK     | BOOL       | HMI soft deadman A (from Test Screen PUSHBUTTON A toggle, D-MAN A tag). ORed with X2 in rung 5.                                |
| C24     | TESTMAN_B_OK     | BOOL       | HMI soft deadman B (from Test Screen PUSHBUTTON B toggle, D-MAN B tag). ORed with X3 in rung 5.                                |
| C25     | OVERCURRENT_TEST | BOOL       | HMI soft overcurrent trigger (Test Screen OVERCURRENT momentary button). ORed with X5 in rung 19.                              |
| C26     | E_STOP_TRIG      | BOOL       | E-stop triggered flag. SET by rung 17 when NOT X4. Shown on Screen 3 EMERGENCY STOP indicator.                                 |

*continued on next page...*

| Address | Name in Ladder             | Type         | Description                                                                                                           |
|---------|----------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------|
| C99     | T4 output                  | BOOL         | Output of T4 off-delay timer (3 s). Active 3 s after CHANGE_SCREEN sets. Not used further in current build.           |
| T0      | Overcurrent off-delay      | OFFDTMR 2 s  | Input: DEADMEN_OK AND (X5 OR C25). Output: JAM_LATCH (C10). Extends jam latch 2 s after overcurrent clears.           |
| T1      | Unjam reverse off-delay    | OFFDTMR 2 s  | Input: JAM_LATCH (C10). Output: UNJAM_R (C13). Drives reverse for 2 s after JAM_LATCH drops.                          |
| T2      | Lockout accumulating timer | TMRA 30 s    | Accumulates total time LOCKOUT_TMR_STRT (C16) is active. T2.DN at 30 s resets C16, stops counting new jams.           |
| T3      | Post-reverse pause         | OFFDTMR 2 s  | Triggered by T1.DN rising edge. Output: PAUSE_MOTOR (C17). Holds Y6 LOW for 2 s after reverse finishes.               |
| T4      | Screen-change timeout      | OFFDTMR 3 s  | Input: CHANGE_SCREEN (C22). After 3 s, T4.DN fires rung 30: resets V100 to 0 (HMI returns from error screen).         |
| CT0     | JAM_DETECT                 | CTU preset 3 | Counts jam events. UP input: JAM_LATCH AND NOT T2.DN. RST input: ERROR_RESET (C18). CT0.Done (N0=3) triggers lockout. |
| V100    | ERROR_SCREEN               | Int (BCD)    | HMI screen selector written by ladder. Rung 29 writes 3 (→ Screen 3), rung 30 writes 0 after T4 (3 s).                |

*continued on next page...*

| Address | Name in Ladder | Type       | Description                                                                                          |
|---------|----------------|------------|------------------------------------------------------------------------------------------------------|
| V2000   | RPM_CTRL       | Int 16     | Speed setpoint (0–1600 counts). Source for rung 2 SCALE → WY0. Written by HMI IncDec buttons.        |
| V3000   | CURRENT        | Int 16     | Scaled motor current (0–140). Output of rung 3 SCALE from WX0. Displayed on Screen 2 CURRENT field.  |
| ST0     | \$FirstScan    | System bit | Do-more system coil HIGH for exactly one scan on power-up. Used in rung 1 to RST C11 (init trigger). |
| ST1     | \$On           | System bit | Do-more system coil always HIGH. Used in rungs 2 and 3 to run SCALE instructions every scan.         |

## 16.7 Appendix G: Test Plan

Test Plan v1.0 comprises a top-down power-on walkthrough followed by ten bottom-up test cases. Each case is exercised against the must/should requirements (M-series).

| ID       | Test case                                                                      |
|----------|--------------------------------------------------------------------------------|
| Top-down | Power-on walkthrough: AC in → PSU → PLC boot → HMI IDLE/READY                  |
| BU-01    | AC power input and fuse distribution                                           |
| BU-02    | 24 VDC PSU output (Mean Well NDR-480-24)                                       |
| BU-03    | PLC rack initialization (DirectLOGIC 205)                                      |
| BU-04    | E-stop hardware safety circuit                                                 |
| BU-05    | Two-hand start (deadman) safety logic                                          |
| BU-06    | Lid interlock (n/a in this build — access guarded mechanically by the ME team) |
| BU-07    | VFD speed command and motor RPM (HY02D211B-T)                                  |
| BU-08    | Overcurrent / overload protection                                              |
| BU-09    | HMI operator interface (EA1-T4CL C-More Micro-Graphic)                         |
| BU-10    | Status indicator lights (power / running / fault)                              |

Validation results to date are summarized in the Testing and Validation and Results sections; load testing under real shredding load is pending mechanical integration.

## 17 Authors and Acknowledgements

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### Disclaimer

This report documents an in-progress senior capstone design. Findings, parameters, and as-built details reflect the current state of the prototype and are subject to revision through final validation.

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*End of rough draft. Comments and questions welcome.*

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